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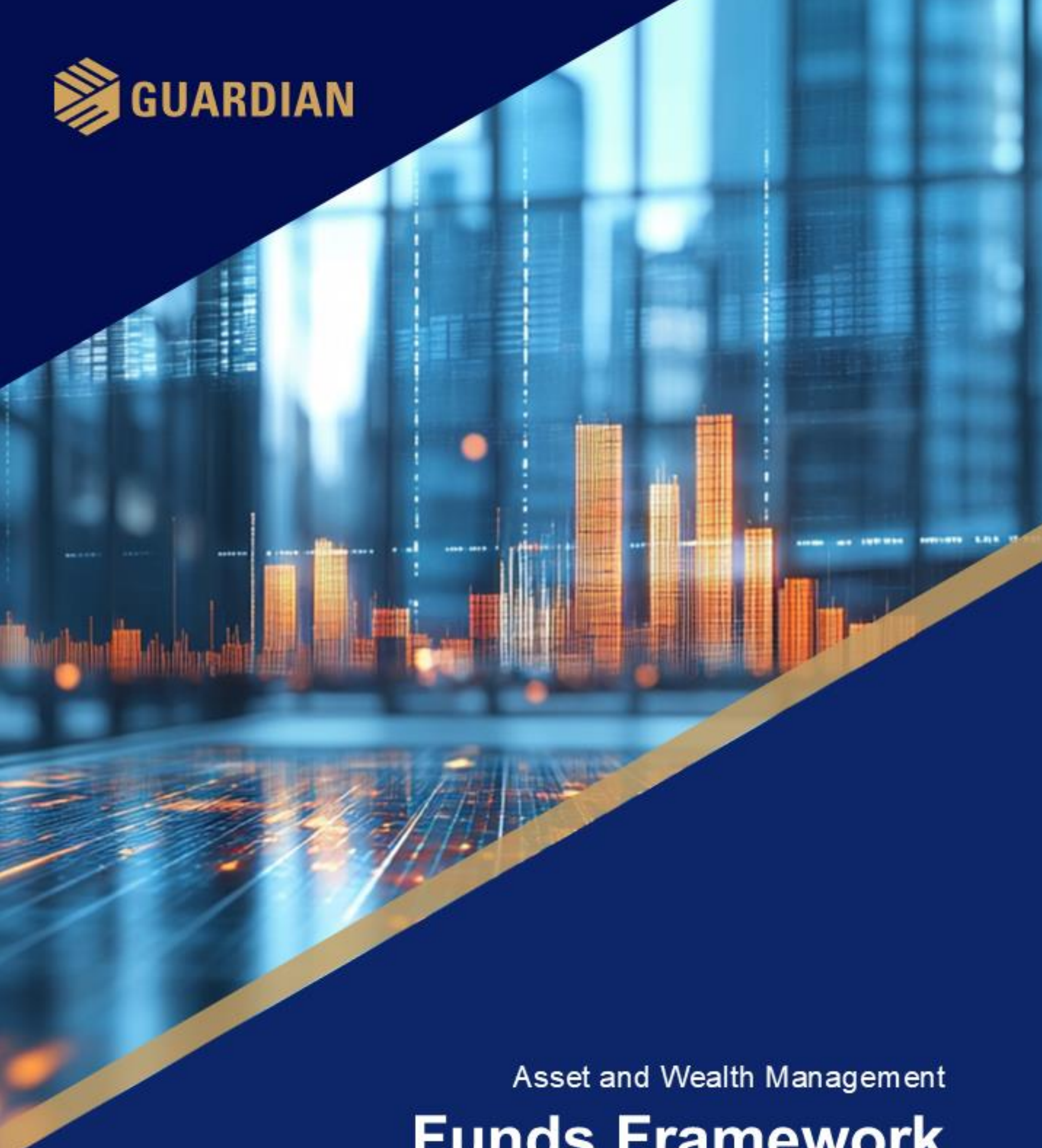

Asset and Wealth Management

Funds Framework

NOVEMBER 2024

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The objective of this report is to highlight the learnings and benefits of tokenisation along with providing an understanding of the risk, technical and operational considerations involved for the fund industry. The document will not cover the areas that are already well established like setting up underlying fund structures such as a unit trust or a variable capital company structure.

Document Version

Version	Date	Author	Rationale
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This is a joint report developed by the Guardian Asset and Wealth Management Industry Group. The development of the report was led by Schroder Investment Management Limited and Franklin Templeton.

The report received additional contributions from organisations listed alphabetically:

- Apollo
- Deutsche Bank
- Financial Conduct Authority (FCA)
- The Hongkong and Shanghai Banking Corporation Limited (HSBC)
- J.P. Morgan
- Monetary Authority of Singapore (MAS)
- SBI Digital Markets
- Standard Chartered
- UBS Asset Management
- United Overseas Bank Limited (UOB)

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1 Introduction

The fund industry is a cornerstone of the global financial landscape, managing trillions of dollars in assets and providing investors with access to diverse investment opportunities. Offering avenues for asset allocation across different asset classes, funds provide diversification of investments and risks for wealth management. Nonetheless, fund operation and distribution are not without its challenges – ongoing expenses, limited transparency, and varying degrees of accessibility remain as hurdles especially for end investors.

A key trend in the fund industry today is the shift towards more liquid and low-cost funds such as index funds, made possible by a combination of investor demand, technology advancements, and regulatory changes. Nevertheless, factors such as market volatility, geopolitical risks, and an evolving regulatory landscape continue to post challenges to the adoption of new technologies and introduction of operational changes. Furthermore, the fund industry functions under stringent regulatory supervision aimed at ensuring transparency and protecting investors. This complexity is heightened when dealing with cross-border fund transactions involving multiple jurisdictions and their respective regulators.

Advancements in digital technologies presents an opportunity to simplify traditional asset management and introduce composable finance[1] throughout the financial industry. Tokenisation has enabled fund managers to re-imagine and modularise the fund lifecycle to better deliver personalisation directly to investors, using records that are stored immutably and securely on distributed ledgers. Despite this, tokenisation initiatives have been implemented in various ways and have remained fragmented across multiple private and isolated networks and initiatives. Thus, to deliver real and widescale benefits in this space, there needs to be standardisation across these networks and initiatives, and close alignment across the industry and regulators.

To this end, Project Guardian has established an Asset and Wealth Management (AWM) industry group, bringing together the combined working knowledge and experience of members across the fund industry. This report introduces the **Guardian Funds Framework (GFF)** as an initial set of non-prescriptive standards and industry best practices for tokenised funds to enable the adoption of asset tokenisation for the fund industry. This report covers the following four areas (a) tokenisation and its applications in the fund lifecycle from origination and issuance, distribution and custody, to secondary market trading; (b) composable technical standards and implementation models; (c) risk considerations; and (d) next steps and future work to further meet the investors' needs.

Through earlier work and pilots in Project Guardian, the AWM members recognise the need to start the journey with a robust and reusable approach to the tokenisation of traditional investment vehicles. Moreover, it is acknowledged that existing fund structures possess limitations, and that there are constraints to the benefits of tokenising these structures as they are currently defined. Consequently, it is important to outline an evolution beyond fund tokenisation to an envisioned future state of a digital asset ecosystem, which maximises the benefits distributed ledger technology to investors and the asset and wealth management industry.

That said, the guidance offered within this document towards initial tokenisation efforts is a critical first step on a path to a fully digital ecosystem. The envisioned future state presents the opportunity to directly address investors' needs with a new, simpler and asset-agnostic operating model. This approach facilitates a fairer distribution of risk, increased efficiency and significant reduction in cost and time required with incorporating new assets and products. Ultimately, it could promise simplification in both technology and regulation frameworks.

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[1] "Composable finance" refers to the construction of financial products, assets and portfolios from independent, inter-operable components, facilitating the reuse of technical code, legal constructs, regulatory rules and operational processes.

2 Background

2.1 Traditional fund industry

The global fund industry manages trillions of dollars in assets, and serves several key roles in the financial sector. They offer avenues for asset allocation across different asset classes, providing investors with opportunities for diversification and potentially reduced risk. The traditional fund management industry caters to a diverse audience from individual investors to institutional entities like pension funds, insurance companies, and charitable foundations. By offering exposure to a range of markets and sectors, they expand investment opportunities for these investors. Some funds specifically cater to long-term goals like savings and retirement planning. Others fuel economic development, with funds like mutual funds, private equity funds, and venture capital funds providing businesses with vital capital for operations and growth. Certain funds like exchange-traded funds (ETFs) provide liquid investment options, while hedge funds offer sophisticated wealth management strategies for high-net-worth individuals.

Funds can be either actively managed or passively managed. Actively managed funds aim to outperform the market index, driven by in-depth analysis and discretionary decisions of fund managers; whereas passive funds replicate the performance of specific indices. Recent trends indicate a shift in demand towards index-tracking and low-cost funds, and the adoption of technology in fund management. This evolution reflects a broader movement towards greater efficiency and cost-effectiveness, as investors increasingly seek transparency and more value for their investment choices. This compels fund managers to innovate and enhance offerings to cater to investor needs and market dynamics.

The industry operates under stringent regulatory oversight to ensure transparency and the protection of investor interests. Regulatory bodies differ by region, such as the Securities and Exchange Commission (SEC) in the U.S., the Financial Conduct Authority (FCA) in the UK, the Monetary Authority of Singapore (MAS) in Singapore, and the European Securities and Markets Authority (ESMA) in Europe.

2.2 Wealth management

Within the wealth management industry, building and managing discretionary portfolios for wealthy individuals is a $5.5 trillion business that helps millions of investors to meet their financial goals[2]. The top wealth managers in this space have built robust and sustainable businesses that have delivered results for their investors. However, their ability to create innovative solutions and realise further efficiencies is constrained by traditional infrastructure platforms that are not built for customisability and interoperability.

Wealth management firms work with clients in a variety of capacities. Their offerings can generally be described as advisory or discretionary.

In an advisory or brokerage relationship, the wealth management firm provides an investor with advice, generally through a series of individual trade or fund recommendations. If the investor agrees, he/she can act on that advice, but the wealth management firm cannot act on the investor’s behalf.

In a discretionary relationship, the investor and portfolio manager (PM) agree on an investment objective, allowing the PM to make ongoing investment decisions on behalf of the investor without requiring further approvals. To provide this service at scale, the chief investment officers (CIOs) at these firms may provide services based on model portfolios which represent collections of assets combined to achieve different risk and return profiles. For example, a balanced ESG-focused portfolio would seek long-term capital appreciation with moderate volatility by investing in strategies across

[2] This is referenced from "The Future of Wealth Management" paper by Apollo and Onyx, J.P. Morgan.

multiple asset types that emphasise sustainability. These model portfolios contain approved investments and recommended weightings by asset class as a template for PMs. An example model portfolio allocation may have 50% in equities, 30% in fixed income and 20% in alternatives.

Public assets such as equities, ETFs, and mutual funds benefit from industry-wide settlement venues, straight-through processing technology, improved real-time access to asset and market data, a broader range of market participants, and therefore, increased liquidity and accessibility. As a result, these assets are well suited to the discretionary model.

In contrast, private assets or alternatives are commonly offered on an advisory basis, where a wealth management firm curates a short list of approved funds from which clients can select and invest. If clients agree to subscribe to these funds, wealth managers guide clients through the process; however, clients ultimately must complete the subscription documents themselves and direct the movement of funds. The result is that the process of subscribing individual investors to private assets has embedded friction, which limits distribution at scale, despite the portfolio enhancing characteristics that alternatives can offer.

3 Problem statements and motivation

3.1 Challenges with existing fund structures

While traditional fund structures have enjoyed widespread adoption, they suffer from well-recognised issues such as high fees where markets are fragmented and funds lack scale, limited transparency and varying degrees of accessibility which can limit their benefits for end investors.

Traditional collective vehicles, unless deployed purely for tax-efficiency, are normally targeted at smaller investors. They democratise investment, giving retail clients access to professional investment management. As a result, they rightly attract substantial attention from regulators to protect those clients from abuse. The consequence is a very heavyweight set of rules, which mandate the roles of a very extensive set of intermediaries and service providers. For example, the delivery of a simple equity Individual Savings Account (ISA) in the UK market can require coordination across as many as 16 regulated entities. This complexity translates into more costs in providing access to end investors.

Principal-traded, open-ended funds are required to commit liquidity to their investors, so that they have the flexibility and choice to redeem at will. In times of market stress, it can be difficult to deliver leading to constraints on trading and in-specie transfer of assets to the redeeming investors.

Underneath the liquidity risk to investors, is a fundamental risk posed by liquidity mismatch between the investment vehicle and its underlying assets. The promise of liquidity from fund structures such as Open-Ended Investment Companies (OEIC), Variable Capital Companies (VCC), Société d'Investissement à Capital Variable (SICAV), and unit trusts, is made, but the assets underlying funds exhibit variable degrees of liquidity. Given the scale of fund investments this can present a systemic risk to the stability of the financial ecosystem under adverse market conditions that cannot be solved simply even with tokenisation.

The costs of fund delivery, and the need to commit asset allocation in favour of liquid assets, both dilute the achievable performance of conventional funds. There is a clear market perception that active funds are expensive, and do not always justify their fees. The market has shifted strongly in favour of passive and exchange traded funds (ETFs), which offer lower fees. While the explicit fees and the number of intermediaries can be lower for ETFs, there are less explicit ways in which the intermediaries (market-makers, authorised participants, brokers, etc.) extract value from the funds: they take value from the investors through arbitrage, spreads and commission. The cost savings can be illusory.

There is a huge choice of funds in the market: risk and return can be sourced from a wide range of asset classes, geographies, strategies and investment styles. However, all investment funds deliver essentially the same product: the net value delta between investment and redemption, plus income, and less the costs and profits of the fund provider.

While clients can choose from the universe of funds available in the market, once invested, there is no choice of exposure, or of value flows. Each investor gets the same risk, the same returns or losses, and the same flows as everyone else invested in the fund. The provider of the fund creates the fund, and promotes it through platforms and distribution channels. The commercial model is essentially product-push, rather than being fully client-driven.

While funds have had success in delivering professional investment to smaller investors, they are no longer the most cost-effective vehicles. Digitalisation presents a transformative opportunity to move strongly to client-driven investment and client-specific solutions in a more cost-efficient manner. As a result, funds in its current form will likely evolve through digitalisation to provide that value proposition to investors.

3.2 Challenges in portfolio management

Managed accounts, based on individualised model portfolios, have grown in popularity in retail markets given their diversification, simplicity, typical goals and risk-orientation, making them appealing to investors with specific objectives in mind. Concurrently, alternative assets have evolved from a small portion of portfolios to a core investment strategy for many investors. However, while the adoption of alternatives has increased greatly, there are still real challenges to their further growth and inclusion in model portfolios.

Adjusting portfolio allocations across asset types (both public and private) requires multiple systems, manual processes and multi-party reconciliation. The process of adjusting portfolio allocations through transaction order management, execution and settlement involves multiple systems and manual intervention by disparate teams. It is not uncommon for PMs to manually calculate trades on spreadsheets and submit trade tickets to operations teams specialising in processing transactions for specific investment vehicles. The operations professionals then interface with the funds' transfer agents or trading venues to submit orders and coordinate cash movement. An extensive reconciliation process across multiple participants is required to ensure investors' cash and investment balances reflect the PM allocation plan. This process regularly occurs for dozens of investment options over tens of thousands of client portfolios.

These friction points also negatively affect client outcomes. Inefficiencies related to receiving cash proceeds from sell orders results in PMs maintaining a cash buffer to ensure they can simultaneously place buy orders. This excess cash is an opportunity cost for the end investor and prevents them from maximising their potential returns.

Distributed ledger technology, and the tokenisation of the underlying assets of funds, offer the real prospect of investment individualisation. The reduction of cost and complexity in 'digital vs digital' transactions, and the ability of self-executing smart contracts to build bespoke portfolio models, together make it feasible for bespoke investment to be delivered to a much wider population of investors.

3.3 Moving from product to client centric approach

With the demography of investors becoming increasingly digitally native, the demand for instant, customisable, cost effective and transparent services increases. The asset and wealth management industry needs to adapt and meet this challenge. Some notable banking providers have already achieved this, highlighting the pressing need for asset managers and banks to adapt, progress and move quicker.

Currently, clients are sold a set of pre-defined fund products with no firm or persistent linkage to their investment objectives. If tokenisation is fully pursued, the situation can be reversed, allowing investment to become a client-driven process.

A potential future state is to enable clients to play a part in the wealth advisory process and be able to co-create a model based on their future financial needs and objectives. This model can be relatively sophisticated, including growth aspirations, loss tolerance, liquidity needs and value flow requirements over time.

In this future state, needs are expressed in a set of tokens that the investor would like to receive, held in the investor's wallet, and exposed to potential providers; providers can then bid to deliver these tokens to the investor. Risk can be with the investor, with the manufacturer, or shared as the parties wish.

Financial advisors will facilitate investor's acceptance of one or more bids, and the winning manufacturer(s) will then mint the tokens that commit the required outcome, and issue them to the investor. So long as willing bidders (or sources) are found for the tokens demanded by the investor, then the tokens issued to the investor will be firmly correlated to the defined outcome objectives. The tokens will trigger and self-execute the defined flows, and reviews will only be necessary when the investor's outcome objectives change.

Adding self-executing flow commitments, is key to deliver more precisely to client needs, and to shift the culture from a product-driven industry to a client-driven one.

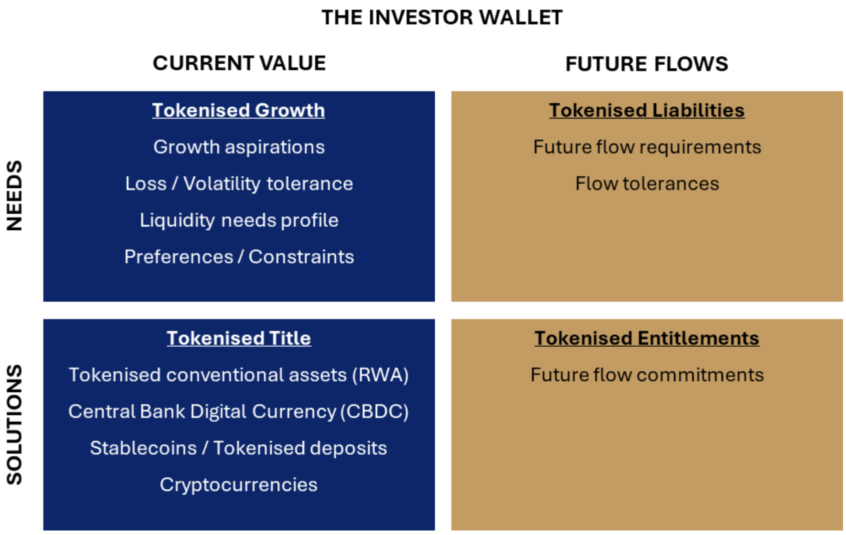

Figure 1: Investor-driven model

3.4 Pathway to realising the benefits of tokenisation

While the benefits of tokenising ownership of conventional vehicles are attractive, there are limits to those benefits. If tokenisation is restricted to digital representation of ownership, while underlying vehicles remain conventional, the full benefits of tokenisation will not be realised. There needs to be a strategic evolution to deploy the advances that distributed ledger technology (DLT) offers for the whole financial ecosystem, with the following three proposed stages over time:

Stage 1: Tokenisation of conventional/traditional funds

The first stage brings DLT into the conventional funds' ecosystem. The tokenisation of conventional funds has been promoted under several global initiatives, including Project Guardian in Singapore and the Asset Management Task Force in the UK. There have been some early successes in tokenised funds, particularly in Money Market Funds (MMFs). However, these early efforts of asset tokenisation have focused on creating digital ways of owning conventional assets. For digital assets to be created, changes are required to be made in its underlying operating models, entities and regulations.

Stage 2: Tokenisation of underlying assets

The second stage is to tokenise the underlying assets and enable efficient trading, registry and settlement, to create customised and bespoke portfolios of tokenised assets for investors. Moving away from the complexity of conventional collective vehicle structures removes layers of cost and intermediation that then enables regulations to also be simplified. This enables composability, in both portfolios and regulations. Composability, supported by the ability of smart contracts to build bespoke portfolio models through self-execution, will facilitate the delivery of tax-efficient, individualised portfolios to investors.

Stage 3: Tokenisation of flows

The third stage is to leverage the self-executing capabilities of smart contracts further, to automate tokens representing entitlement to value flows. This builds on the tokenisation of funds and conventional assets delivered in the previous two stages. With concerted efforts from the industry, this will enable financial assets and products to be composed from underlying flow commitments facilitating a single, self-executing operating model across all financial assets. This is the fundamental underpinning of simplification in process, technology and regulation.

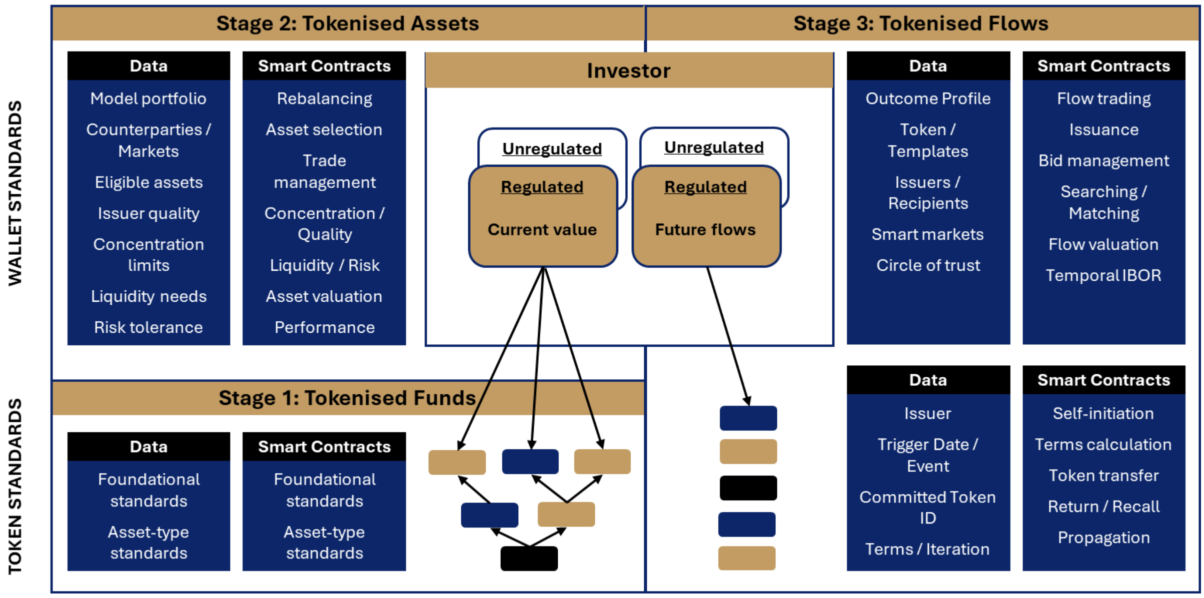

Figure 2: Evolutionary stages to realise full benefits of tokenisation

This report will focus on the key features and considerations in the first stage: fund tokenisation.

Tokenised investment vehicle – the first digital step forward

The first stage of the asset and wealth tokenisation journey begins with a Tokenised Investment Vehicle (TIV): a tokenised conventional/traditional fund. The token represents title to a proportional share of the fund's assets, enabling streamlined and direct trading on a DLT infrastructure. This can significantly cut down the time and complexity related to the activities in the fund lifecycle (e.g. origination, issuance, distribution). Simplifying the investment journey would also mean reducing the intermediaries often required in conventional transaction processes. This ultimately allows fund managers to provide investors with greater accessibility, transparency, and efficiency.

For fund managers considering tokenising a fund, starting from the ground up can be daunting. As with any pioneering developments in finance, setting up a TIV presents its own unique set of challenges, including regulatory clarity, technical standards, risk factors, legal and tax implications. The subsequent sections in this report will dive deeper into these areas covering the tokenised fund lifecycle and technical standards.

3.4.1 Tokenisation in portfolio management

Tokenisation has the potential to harmonise the treatment of public and private assets in portfolio management, creating significant value for asset and wealth managers and investors.

A common operating model across financial assets and products can be supported by a single infrastructure. This same infrastructure will support not just existing assets and products, but any financial asset or product that is defined in a compatible form. The self-execution of flow commitments eliminates much of the current operational burden of financial product delivery. Much of asset-servicing and settlement management dissolves into the common, self-executing operating model.

Creating new, connected infrastructure can streamline, simplify and normalise the investment and operational processes related to building and managing portfolios. This technology change could revolutionise the asset and wealth management business by fundamentally redesigning how portfolios are accessed, assembled and managed. Over time, this will provide investors with greater access to higher quality investment portfolios in a more efficient and personalised manner. Achieving this could evolve the discretionary portfolio management business by enabling the seamless deployment and automatic rebalancing of a large number of portfolios. Additionally, it could make the case for including alternatives in discretionary model portfolios significantly stronger, as operating standards between traditional assets and alternatives converge.

Successful implementation in the near term would be most beneficial to the group of individual investors who are currently able to invest in alternatives (but may be under-allocated). Examples of these investors include qualified purchasers, accredited investors, and knowledgeable employees of asset managers. They could find it easier to initiate or increase their portfolio allocation to alternatives, which could grow the demand for alternatives without any change to investor suitability requirements.

As this progresses through the stages recommended in this report, the number of entities required to deliver financial products reduces, and the roles change to deliver benefit more directly to clients.

Current State	Stage 1: Tokenised Funds	Stage 2: Tokenised Assets	Stage 3: Tokenised Flows
Financial adviser	Financial adviser	Financial adviser	Financial adviser
Tax-wrapper provider	Tax-wrapper provider	Tax-wrapper provider	Tax-wrapper provider
Fund entity	Fund entity	**Asset manager**	**Asset / Liability manager**
Fund manager	Fund manager	**Digital custodian**	Digital custodian
Platform	Platform	Asset servicer	**Liquidity provider**
Distributor	Distributor	Cash token issuer	**Valuation / ALM verifier**
Custodian	Custodian	**Token order manager**	**Token / Flow marketplace**
Depository	Depository	**Valuation service**	Settlement optimiser
Asset servicer	Asset servicer	**Token marketplace**	
Payment bank	**Cash token issuer**	Broker / dealer	
Transfer agent	**Transfer agent**	**Settlement optimiser**	
Fund accountant	Fund accountant	**DL – Settlement manager**	
Manco / AIFM / ACD	Manco / AIFM / ACD	Asset pool custodian	
Auditor	Auditor	Asset token issuer	
CASS auditor	CASS auditor		
Legal adviser	Legal adviser		
Market infrastructure	Market infrastructure		
Broker / dealer	Broker / dealer		
Clearing house / CCP	Clearing house / CCP		
Settlement utility	Settlement utility		
	Asset pool custodian		
	Asset token issuer		

Figure 3: Evolution of roles

3.4.2 Automation and efficiency improvements

Leveraging smart contracts to represent and record the ownership of assets could collapse the PM and operations roles, enabling portfolios to be deployed and rebalanced programmatically at scale. The presence of cash and fund ownership records on a shared ledger, combined with smart contract-enabled trade execution, could limit the need for costly reconciliations and occurrence of trade errors. At scale, the time and cost savings could allow wealth managers to reinvest in research and clientfacing services, including client education. Efficiency benefits could accrue to others in the ecosystem, including asset managers, fund administrators, and other service providers.

From an investor perspective, eliminating these friction points could enable PMs to be fully invested more consistently, meaning their portfolios would experience less cash drag. Assuming the average PM holds ~3% cash and a balanced portfolio could generate ~8% over cash in the long-term, the net result to a client is a ~24bps reduction in costs[3].

Other anticipated benefits include:

- **Potential liquidity improvement:** Today, selling alternatives holding on a secondary market is typically a manual process negotiated on a bilateral basis. This can become problematic for individual investors, as oftentimes their holdings are not large enough to attract buyers, given the time and paperwork required to complete a transaction. Simplifying these operational processes using smart contracts, coupled with alternative liquidity methods, such as the automated netting of subscriptions and redemptions, could add additional liquidity levers.
- **Enhanced investment outcomes through alternative investments:** Streamlined processes and enhanced liquidity could allow alternatives to be included in model portfolios and improve expected returns for investors and/or reduce volatility. It may also be possible to automatically rebalance portfolios based on changes to model portfolios, which could minimise deviations from target asset allocations, resulting in portfolios that align better with their optimal state.
- **Combining the efficiency of robo-advisory with the alpha of active management:** Automated portfolio construction and management could provide a streamlined experience similar to robo-advisory offerings, but with a dedicated PM and higher potential returns through three sources of alpha: a) the inclusion of alternatives; b) manager due diligence on active strategies (e.g. identifying a top large cap growth fund); and c) setting top-down asset class allocations based on CIO macro insights.
- **Flexibility and broader access:** Leveraging interoperability solutions to connect distinct DLT networks could provide access to tokenised investments across disparate chains, allowing PMs to build holistic solutions with the inclusion of these investment opportunities, which otherwise might not be accessible.

[3] This is referenced from "The Future of Wealth Management" paper by Apollo and Onyx, J.P. Morgan.

4 Tokenised fund lifecycle

4.1 The traditional fund lifecycle and its key phases

The traditional fund lifecycle consists of two key phases: origination and issuance, and distribution. As a general example, at origination and issuance, it begins with the 'Fund Design' phase, where the investment strategy is formulated, including decisions on asset class, regions, sectors, and more. Once the concept is established, the 'Fund Setup' begins to lay the legal and operational foundation, which includes regulatory compliances and approval from relevant authorities. Upon legal sanction, the fund enters the distribution phase where it is open to receive investments. A distributor facilitates access, allowing investors to subscribe to or redeem fund units.

Money received for investment is strategically invested into assets and rebalanced as required to maintain alignment to the strategy. The underlying assets of the fund are held with a custodian.

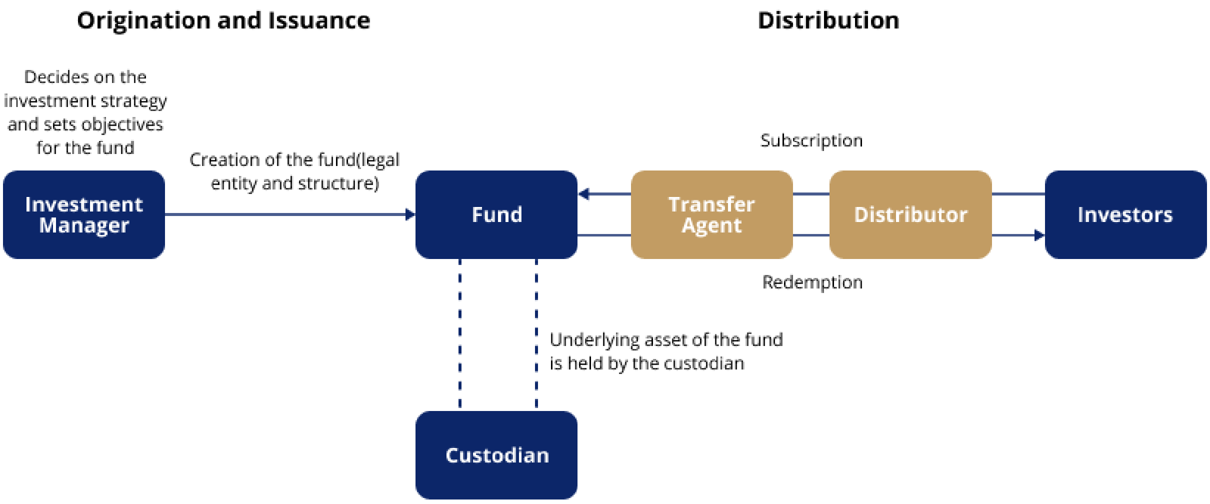

Figure 4: Traditional fund lifecycle

The following section will describe the role of tokenisation in the fund lifecycle across these two key phases, and how this can potentially lead to the opening of secondary markets with tokenisation.

4.2 Origination and issuance

The origination and issuance of a TIV involves several key steps to create a digital representation of title to an investment product. The fund manager determines the approach to achieve the fund's objectives. It includes decisions around asset allocation, sectors to invest in, active vs passive management, geographical focus, and more. Objectives are the specific, measurable goals that the fund seeks to achieve over a set period. This could include capital preservation, growth, income generation, outperforming a specific benchmark, or a combination.

A simplified outline of this process goes as follows:

Step 1: Asset selection. Depending on the investment strategy, the assets could be public assets such as equities and bonds, or alternatives like real estate and private loans.

Step 2: Fund creation and legal compliance. This step involves the selection of the desired fund structure and the creation of a new legal entity for the fund, aligning with existing legal and regulatory frameworks.

Step 3: Smart contracts. Smart contracts are programmed to execute functions that may include distributing income and ensuring compliance with regulatory requirements.

Step 4: Token creation. The tokens are then issued through the subscription process. Subsequently, the trading of tokens may occur on an exchange where the tokens are listed, or with the issuer in a principal-traded fund.

Step 5: Lifecycle management. This includes the operational tasks post-issuance, like additional issuance or burning of tokens, corporate actions, managing investors rights, review of current distribution vs product design goals and its target market etc.

For a TIV, the records on the DLT may represent the authoritative fund registry entirely where regulation permits. As mentioned in the previous section, depending on the extent of tokenisation, the underlying investment assets may be tokenised, and portfolio management and operational processes may be on the DLT to enhance efficiency. The legal and operational fund structures may vary depending on the level of implementation.

The services provided by the participants in the fund ecosystem will evolve as activities become automated through the use of smart contracts.

4.3 Distribution

A distributor plays a pivotal role in connecting investors with appropriate financial products. Their responsibilities include promoting and selling financial products, providing tailored investment advice, ensuring regulatory compliance around investor eligibility, and supporting clients throughout the investment process to help them to achieve their financial goals. Emerging channels like robo-advisors, which are increasingly popular among younger demographics, are also adding a new dimension for traditional distributors such as banks, insurance companies, and financial advisors.

4.3.1 Complex asset-servicing ecosystem

Financial products distributed by intermediaries are often managed by multiple asset managers, each utilising their own network of asset servicing partners, including transfer agents, calculation agents, and paying agents. The complexity is further compounded by the diverse range of investment products, each associated with different asset classes and servicing requirements. This complexity raises a pertinent question: Can tokenisation streamline operational processes, enabling distributors to offer a broader range of products while reducing the complexities associated with managing scale and diversity in the product set?

4.3.2 Intensifying customer demand for shorter response times

Today's consumers expect instant gratification, as evidenced by the rise of on-demand services in streaming and food delivery. However, the lifecycle of asset and wealth management products often involves delays, with activities such as product purchases, payouts, and updates on corporate actions taking several days to weeks. For instance, the lifecycle of a structured note from subscription to settlement can span 10 to 14 days; transfers of client assets between fund platforms can take 4 to 6 weeks. These delays are exacerbated by the presence of multiple intermediaries, time zone differences, and system cut-off times. Digitalising assets and incorporating smart contracts on a DLT network could shorten these timelines and enhance efficiency.

4.3.3 Inaccurate information transmission

The intricate nature of asset servicing often results in challenges to timely and accurate transmission of information. Asset and wealth management workflows, particularly for alternative investments, are siloed and involve multiple intermediaries. This fragmentation makes it difficult for distributors to consolidate and relay information to investors efficiently. Each intermediary expects information in a certain format or converts information to the formats their systems accept; they convert outgoing information to other formats expected by the world outside to them.

The process is subject to errors, breaks and inaccuracies, and incurs expensive technology/operational changes when a party in the workflow makes a change to their transmitted information or their way of operating. As an example, when a corporate actions event such as rights issues or voting occurs with a short notice to respond, the pressure to communicate timely and accurately increases the risk of errors. DLT, which provides a single, immutable record accessible to all parties, could potentially resolve these issues by eliminating discrepancies in information transmission. With timely access to accurate information, the immutable record would serve as a foundation for building investor confidence, especially in alternatives that typically offer limited transparency.

4.4 Secondary market

Secondary markets offer several advantages for investors, opening up the possibility of secondary pools of liquidity and collateralisation. By creating a marketplace for fund tokens, secondary markets could reduce price opacity and reduce information asymmetry, where timely, good quality disclosures are made to the market, especially for private asset investments.

For fund managers, secondary markets can enhance fundraising efforts by demonstrating liquidity and attracting a broader investor base. By tokenising fund units, investors can trade across more avenues in the secondary market, which creates more options for investors to enter and exit investment funds, including alternatives.

This section outlines potential secondary market use cases and ecosystem considerations which may have implications to a TIV. These benefits are not dissimilar to those seen in the growth of ETF markets.

- **Enhanced liquidity:** Secondary markets provide investors with the ability to sell their fund tokens at any time, unlike traditional principal-traded funds, which are bound by dealing cycles and fund holidays. Secondary markets can unlock liquidity that is typically absent in alternate investments. This can be particularly beneficial during periods of market volatility or when investors need to rebalance their portfolios.
- **Price discovery:** The trading of fund tokens on a secondary market can reduce information asymmetry between buyers and sellers, as all transactions are recorded in a transparent and immutable manner. This can lead to more efficient price discovery.
- **Access to a wider range of investors:** Secondary markets can attract a broader investor base, including retail investors, family offices, and institutions that may not have traditionally invested in such assets.
- **Market intelligence:** By observing trading activity on the secondary market, fund managers can gain valuable insights into investor sentiment and asset valuations.
- **Fractional ownership**: Investors can buy and sell smaller portions of assets, making high-value investments more accessible.

However, there are challenges to adoption:

- **Lack of regulatory certainty:** The regulatory landscape for tokenised securities and secondary markets is still evolving, creating uncertainty for market participants.
- **Custodial and security risks:** Ensuring the safekeeping of digital assets and protecting against cyber threats is crucial for the development of secondary markets.
- **Market infrastructure:** Building the necessary infrastructure, including trading platforms, clearing and settlement systems, and custody solutions, requires significant investment and coordination.
- **Investor education:** Many investors are unfamiliar with tokenisation and secondary markets, necessitating investor education and awareness campaigns.

- **Market liquidity**: For secondary markets to function effectively, they need a certain level of liquidity. If insufficient investors are interested in buying and selling TIVs, the market for them may suffer from low liquidity, affecting price discovery. Listed closed ended funds often trade at significant discount or premium to fund NAV.
- **TIV Liquidity:** While secondary markets offer the potential for increased liquidity, achieving sufficient trading volume to establish a robust market can be challenging for some fund types that may contain strict lock-up periods, infrequent redemption cycles and illiquid underlying assets.
- **Market fragmentation**: If the same TIV is traded on multiple networks and platforms, this could fragment liquidity and complicate price discovery.
- **Valuation:** Digital ledgers could provide a mechanism to disseminate and record the issuance of disclosures on events relating to underlying assets and amendments to fund valuations, enabling secondary markets to operate in a more effective and orderly manner.

4.5 Custody and security

When a fund is tokenised, an investor will hold the fund tokens in a digital wallet represented by a pair of keys. The investor's private key will be secured in a digital wallet. A choice would typically be to elect whether a custodian holds the private keys or investors hold their own private keys.

- **Digital Asset Custodians:** These institutions provide secure storage and management of digital assets, including fund tokens. They play a crucial role in protecting investor funds and preventing unauthorised access. Where investors do not hold keys directly, a digital wallet custodian will undertake responsibilities including holding investors' private keys, and monitoring the process to approve transactions.
- **Security Protocols:** Robust security measures, such as encryption, multi-factor authentication, and cold storage, are essential to safeguard digital assets from cyber threats.

4.5.1 Considerations

A robust ecosystem is essential for the successful operation of secondary markets for tokenised funds. This ecosystem comprises several key components:

Venues

In theory, fund tokens can be listed on any agreed trade execution venue to create a secondary market in the tokens, thereby increasing the liquidity of the tokenised fund. The venue, whether it is an exchange, private bank, wholesale bank or another platform, will need to provide clients a platform that combines product disclosure, KYC and marketing resources to capture market share and ensure appropriate investor education.

- **Digital Exchanges:** These platforms facilitate the buying and selling of tokenised funds, providing order matching, price discovery, and trade execution services.
- **Order Books:** Order books are electronic records of buy and sell orders for tokenised funds, allowing market participants to view available liquidity and prices.
- **Marketing & Educational Resources:** Providing investors with information about tokenisation, secondary markets, and associated risks is crucial for building trust and confidence.

Platform connectivity

Connectivity between tokenisation platforms presents investors increased trading options. However, when transactions traverse borders, the complexity of maintaining compliance is non-trivial.

Tokenisation platforms are subject to the regulations of the jurisdictions in which they operate. Variations in compliance requirements across countries can make coordination and interoperability difficult, posing a challenge for platforms seeking to develop cross-border compatibility. In addition, interoperability must not compromise KYC and AML obligations. Differences in these rules across jurisdictions may cause divergence in verifying and standardising due diligence procedures.

Technically, a secondary market ecosystem will need a network to connect issuers and distribution venues where the tokens can be securely replicated or transferred. This could be accomplished by establishing interoperability, or exchanges connecting directly to ensure integration between different platforms. Alternatively, it could be facilitated through intermediaries with custodians or other service providers providing connectivity between multiple DLT networks or venues.

Exchange of value

For DLT network transactions to settle efficiently, tokenised cash must be exchanged for tokenised funds. The exchange of value on a DLT is immediate via atomic settlement – there is no intermediary. If either the cash or the token is not delivered, the transaction will not happen, removing the counterparty risk that exists in bilateral transactions today. Therefore, the digital wallet of the tokenised fund and cash must be fully funded prior to settlement, or the transaction will fail.

As traditional fiat currency is not transferrable on DLT networks, the use of payment tokens is necessitated. This workaround can be inelegant, and the attributed pre-funding expense might offset the advantages of tokenisation. In many cases, commercial bank money is available as a settlement mechanism on DLT networks, while Central Bank Digital Currencies (CBDCs) are also being tested with many jurisdictions today. Developments on digital money point to potential further innovation happening over the next three to five years, which may result in one or more common settlement assets for tokenised funds.

4.5.2 Alternative assets

Alternative assets, such as private credit and equity, are particularly suitable for tokenisation due to their manual and bespoke nature. Tokenisation can streamline operations, improve liquidity, and enable smaller ticket sizes, making these investments more accessible to individual investors. However, creating and sustaining liquidity requires aligned incentives across stakeholders, including fund managers, distributors, and investors.

- **Fund-of-funds:** Tokenisation can efficiently enable the creation of fund-of-funds, allowing investors to diversify their private equity exposure across multiple underlying funds.
- **GP-led secondaries:** General partners (GPs) can utilise tokenisation to facilitate secondary transactions, providing liquidity to limited partners (LPs) and unlocking value for the fund.
- **LP secondaries:** Tokenisation can streamline the process for LPs seeking to sell their interests in private equity funds, providing access to a wider pool of potential buyers.
- **Real Estate Investment Trusts (REITs):** Tokenising REIT shares can enhance liquidity and provide investors with greater flexibility.
- **Commercial Mortgage-Backed Securities (CMBS):** Tokenisation can create a more efficient market for CMBS, allowing for the securitisation of larger and more diverse loan pools.
- **Loan syndication:** Tokenisation can facilitate and accelerate the syndication of loans, enabling a wider distribution of credit risk.
- **Structured products:** Complex financial instruments, such as collateralised debt obligations (CDOs), can be tokenised to enhance liquidity and transparency.

4.5.3 Compliance with jurisdictional regulations

Different jurisdictions impose varying guidelines and regulations on digital assets. At the same time, tokenisation does not change the legal nature or status of the underlying asset. As a principle, the same regulations governing the underlying asset should generally apply to its tokenised form, with additional risk mitigating measures introduced to address the specific technological aspects of tokenisation.

For tokenised products intended for distribution, distributors must consider jurisdictional requirements, including:

- **Licensing and regulatory requirements:** Distributors must adhere to licensing and regulatory frameworks based on the activities involving tokenised products and related payment tokens throughout the product lifecycle.
- **Disclosure requirements:** Distributors may be required to provide clear information to investors on any additional risks involved. This includes (non-exhaustively) details about the tokenisation model and token issuer, DLT type, business continuity plans, smart contract audits, and custodial arrangements.
- **Selling restrictions:** Regulations may impose restrictions on the sale of tokenised products based on investor type and/or domicile.
- **Data classification and processing:** Compliance must be maintained with data classification and processing regulations for customer and personal data.
- **AML/CFT requirements:** Distributors and wallet providers must address anti-money laundering (AML) and counter-financing of terrorism (CFT) requirements specific to tokenised products, including travel rule and on-chain screening.

Options for custodial or non-custodial wallets

Depending on the level of sophistication of the customer and their preference, they may choose either custodial or non-custodial wallets. A distributor can either hold the assets in an omnibus wallet (where all assets are held in one wallet) or open a wallet for each client. Customers using custodial wallets interact with their assets through existing channels like their banking applications. On the other hand, those using non-custodial or self-custodied wallets manage their assets through their own digital wallet. At this point, any preference towards self-custody will only be seen in sophisticated Web3 enthusiasts.

Tokens liability and shared responsibility model

Potential risks include data loss, security breaches, loss of keys, smart contract failures, and environmental failures. It is important to note that existing technologies also have similar inherent risks. Who should bear the liabilities if the private keys to the wallet are compromised, misplaced or stolen?

If the assets are held in a centralised custodial wallet maintained by a distributor, the distributor would likely take full responsibility, and the losses would not be sustained by the customers, subject to prevailing legislation on prudential resources or insurance governing such firms.

However, if the assets are held in non-custodial or self-custodied wallets, establishing responsibility and liability for a loss of control and possession of the assets will require more effort. For example, if the tokens are issued by a smart contract and the operator of the smart contract issued a “burn” or “force transfer” command that no one else was pre-notified of, or has the control to block, the operator would look to be the responsible party. However, should this event be anticipated, and a third party has already been given this responsibility to block unusual commands via a multi-signature key, then the picture is less clear.

If the owner of the self-custodied wallets lost or compromised the wallet key, effectively losing the tokens, then a shared responsibility model could be beneficial to pave trust-saving solutions. This would wrap in the wallet provider, the wallet owner and the operator of the smart contract that issued the token.

One possible solution is for the allocation of liabilities (i.e. shared responsibility) among all token parties to be clearly articulated in the product's legal documentation. This information is typically included in the terms and conditions established by issuers and/or distributors. These terms can be categorised into two main types:

- **Commercial terms:** Some terms and conditions are driven by commercial considerations and are therefore contractual in nature. These provisions reflect the agreements and obligations negotiated between the parties involved.
- **Regulatory terms:** Other terms and conditions are mandated by regulatory requirements imposed on issuers, appointed service providers, and/or distributors, depending on the jurisdiction of issuance and distribution. These terms ensure compliance with local regulations and standards.

A complementary solution is also to implement an acceptable due process to reinstate the lost tokens. This could be through a master override, to mint new ones after determining that the "lost" tokens are inoperable (and destroyed). Clearly, this process would need to be established with regulatory and legal support.

4.6 Asset servicing

Asset servicing refers to the operational support of investment funds across its lifecycle activities including custody, valuations, accounting, compliance reporting and transaction processing like dividend or income distribution. A pivotal role of an asset servicer is to ensure the availability and readiness of quality accurate information for asset managers, investors and regulators in the four important books and records:

- **Investor register**: The register is an investor holding record that consolidates and identifies the ownership of units or shares held by investors. These records include information like the investor's identity, number of units or shares held, purchase, redemption value, value dates, and account balances. Dividend or income distribution to investors rely on these records and therefore regulatory and legal recognition of on-chain records is a necessary condition.
- **Cash register**: Cash balances available for investments by the asset manager and due to investors as a result of redemption requests, to facilitate high quality, holistic cash forecasts for asset managers.
- **Fund Net Asset Valuation (NAV)**: Fund accounting and valuation record that ensures accurate NAV of investors' holdings.
- **Security master**: Custody book of record details the assets and cash that have been settled and delivered to the custodian bank for the purpose of safeguarding assets invested by the fund, that are ultimately owned by the investors.

Some of these records are required by investors for financial and tax reporting while others are used by asset managers for internal and regulatory reporting.

While the asset servicing function covers the four areas above, this section will cover only the subscription and redemption activities, while touching briefly on digital identity, the importance of asset interoperability, standardisation and anti-money laundering.

4.6.1 Subscription and redemption

Subscriptions, redemptions and income/dividend distributions are events that change the investor holding records. For tokenised funds, investors may have an additional option to subscribe to and redeem from on-chain funds using digital currencies, in addition to fiat. Table 1 below shows some combinations of digital currencies (e.g. tokenised deposits, central bank digital currencies, or stablecoins) and fiat currencies that can be used in either the subscription and/or the redemption leg.

This offers unique benefits to investors and asset managers, including faster time-to-market, lower cost and higher accessibility. However, it is not without its challenges.

These benefits and challenges arise when existing workflows and processes change, primarily driven by implementation of the smart contract. In turn, the deployment of smart contracts are driven by business decisions to employ fiat and/or digital currency within the transaction. This impacts the level of programmability required, the presence of digital identities to automate the flows, and the convergence of DLT concepts with traditional workflows. The resulting implications can range from new automation efficiencies to operating model changes for differentiated competitiveness.

Table 1 and Table 2 highlight potential models for on-chain subscription and redemption models, determined by the type of currency used. The type of fiat and/or digital currency used can have an implication on the level of differences in operating and/or workflows from today's traditional technology and operational parameters, noting that most subscriptions and redemptions via fiat are currently processed in batches rather than in real-time.

Investor	Model 1	Model 2	Model 3
Subscription	Fiat (e.g. direct debit, instant payment)	Fiat (e.g. direct debit, instant payment)	Digital currency (e.g. tokenised deposits, stablecoin, CBDC)
Redemption	Fiat (e.g. direct credit, instant transfer)	Digital currency (e.g. tokenised deposits, stablecoin, CBDC)	Digital currency (e.g. tokenised deposits, stablecoin, CBDC)
Potential impact	No change to traditional workflows	Asset manager's cashflow projection unchanged. Investor receives monies faster.	Cashflow projection, unit capital determination faster. Investor receives fund unit tokens and monies faster.

Table 1: Fiat or digital currency for subscription and redemption

Investor	Model 1	Model 2	Model 3
Subscription	No change	No change	Smart contracts can automate digital currency transfers for subscription
Redemption	No change	Smart contracts can automate digital currency transfers for redemption	Smart contracts can automate digital currency transfers for redemption
Potential impact	No change to traditional money transfer workflows. Smart contracts can enable these workflows to be automated.	Manual reconciliations still required. For example, subscription bank accounts may need to be mapped to wallet addresses. Cashflow forecast remains constrained if time lag between subscription (e.g. next day) and redemption (e.g. real time) is significant.	New speed and increased transparency of money movements for better deployment. However, new technology mitigants may be needed if smart contracts are used.

Table 2: Smart contracts in the automation of subscription and redemption

Smart contracts can enable the automated transfers of digital currencies on the DLT network out of/into the beneficiaries wallets upon ensuring requirements are met in subscription and/or redemption processes. Although smart contracts can also be implemented with traditional workflows, benefits such as instant settlement can only be derived when digital assets and money are transacted. This creates an opportunity for asset managers to operate a relatively autonomous 24/7 on-chain operation to facilitate investor cash transactions.

Type of fund valuation	Tokenised fund token with $1 par value	Tokenised fund token with dynamic NAV
Subscriptions before daily cut-off time	Fund tokens can be issued immediately	Fund tokens confirmed on T+1 based on NAV on end of T-date
Redemptions before daily cut-off time	Digital money can be paid out immediately	Digital money will be paid out on T+1 based on NAV on T-date* **Digital money may be paid out immediately with off-market hours funding by liquidity provider*
Redemptions on 24/7 basis	Pay out based on par value	Pay out based on agreed investor terms and conditions regarding the NAV to be based

Table 3: Subscription and redemption of tokenised funds with fixed and dynamic NAV

Fund Type	Tokenised fund with underlying 100% traditional investments	Tokenised fund with mix of underlying traditional + digitally native investments	Digital fund with underlying 100% digitally native investments
Dividends/Income ***Asset entitlement that needs to be distributed / accrued. Impacts NAV.***	Fiat is credited to bank accounts OR Fiat converted into digital currency that is sent to wallets	Fiat is credited to bank accounts OR Fiat converted into digital currency that is sent to wallets	Fiat is credited to bank accounts OR Fiat converted into digital currency that is sent to wallets
Mandatory corporate actions	Follow traditional channels for execution	Mix of traditional and potentially digital channels	Potentially digital channels with smart contracts
Voluntary corporate actions ***Information or asset entitlements e.g., share split, buy-back that need to be actioned***	Follow traditional channels for execution	Mix of traditional and potentially digital channels with smart contracts	Potentially digital channels with smart contracts

Table 4: Managing Entitlements

Table 4 above highlights the impact on dividend distribution and corporate actions for funds with underlying traditional investments, a mix of traditional and digitally native investments, and fully digital funds with digitally native investments. While it may take more time, with increasing development and acceptance of digital currencies, it will likely become acceptable to investors for income and distributions, with further benefit accruing at that point.

4.6.2 Anti-money laundering and governance

The need to detect and prevent illicit activities for tokenised funds continues to be paramount as the fund industry moves towards digitalisation. Introducing guardrails for the subscription, redemption, and income distribution of tokenised funds will be necessary to mitigate such illicit activities as market manipulation, money laundering and other cyber-related crime.

In a decentralised world, verifying the identity of the counterparty or an investor will require coordination across the broader funds' ecosystem. One solution is to use digital identity with an implementation of verifiable credentials to verify parties and their transactions across networks. Through this, investors and counterparties can be identified across different platforms and potentially across jurisdictions, bringing more transparency to their transactions.

On top of digital identity, services such as financial screening tools can further strengthen these operations. Such tools allow for wallet screening and the monitoring of investor activities, enabling the identification of transactions associated with illicit activities in real-time. By combining advanced monitoring solutions and digital identity, fund managers can ensure continuous oversight of funds, enhance compliance with regulatory standards, and safeguard the integrity of the fund industry.

4.6.3 Digital identity in asset servicing

Digital identity or decentralised identifier (DiD) is a form of on-chain digital identity that is unique, non-transferable, and privacy preserving. It can be conveniently used to facilitate investor identification and suitability checks, recordkeeping and distribution of entitlements without the need for a central coordinating party. This allows for the identification of transacting parties across interoperable networks and enables operations to be streamlined across subscription/redemption, income distribution, information sharing and anti-money laundering efforts.

Digital identity can be implemented through smart contract issued cryptographic-based tokens in combination with other authentication methods. More importantly, it can enable an automated and seamless client experience, efficient asset servicing and better on-chain governance and regulatory compliance.

Through whitelisting or verifiable credentials, it ensures that only sufficiently verified investors can hold and trade fund tokens. For example, investors and intermediaries that have been identified and issued digital identity tokens can be quickly authenticated and verified across different networks without requiring confidential data to be stored or processed on-chain. This reduces friction in KYC and investor-suitability processes, while enabling more efficient subscriptions and redemptions.

It can facilitate secure and personalised information distribution such as fund performance updates, regulatory notices and investor reports that are delivered direct to the required wallet addresses. In income distribution, digital identity can ensure accurate distribution, by securely linking investor identities to wallet addresses across different DLT networks. For example, income from a tokenised money market fund can be automatically reinvested into the fund, with the additional fund tokens distributed to the investors based on digital identity tokens.

With digital identities, smart contracts can be used to verify identities to ensure payouts, or to withhold payouts, should compliance flags be raised for that identity. This prevents unauthorised recipients and raises trust that identified beneficiaries are the ones who are receiving the entitlements. While it is a catalyst for efficiency, it does not replace well-designed workflows that need to address the specific fund deployment situation.

Funds with underlying public traditional securities, alongside tokenised / digital assets would require operational considerations for both traditional and digital workflows, in order to manage dividend and other corporate action events.

List of digital identity roles in asset servicing		
Know-Your-Customer	Investor suitability	Sanctions filtering
Anti-Money Laundering	Payments	Whitelisting
Information distribution	Asset ownership evidence	Billing and fees

Table 5: List of Digital Identity Roles in Asset Servicing

Validators performing settlements can also be assigned with digital identities as a method of whitelisting. For example, their identity could include their IP addresses, to avoid whitelisting validators from sanctioned countries.

4.6.4 Other asset services

Asset servicers provide a range of services of which could be improved with DLT. However, these capabilities require customisation for compatibility with existing infrastructure today. Thus, its benefits may not be fully realised until fund lifecycles can be executed on DLT networks, which provides end-to-end processing and management of the fund's lifecycle within the DLT environment. Further, infrastructure will need to be more interoperable, and widely adopted for tokenised funds to scale. That said, there still are areas where smart contracts and DLT could be leveraged for more automation and efficiency today.

One such area is fund administration – leveraging smart contracts and distributed ledgers to capture expenses and income for NAV processing of a fund. Asset servicers process fund NAV off-chain for most fund types today. NAV processing includes a range of activities from new instrument set-ups by portfolio managers, new valuation and accounting set-ups, to the calculation of fund expenses as well as pulling market data from various sources. While oracles could receive market data to facilitate on-chain processing, accounting and valuation set-ups for new instruments across asset classes demand a comprehensive range of functionalities that are time sensitive. Production of fund performance, attribution data and suchlike reports are also currently linked/integrated to NAV, corporate actions, cash and other sources of data that traditional IT systems can effectively perform.

Custody can be multi-layered, with underlying assets and digital funds stored in smart contracts, and across networks. It can also be safekept in "decentralised" ways via investors own wallets, or centralised via wallet-account structures consolidated by intermediaries. However, beyond the safekeeping of the wallet private keys, it is crucial that the custodian is able to exercise possession and control of the digital assets, digital fund or fund unit tokens if these tokens are under custody too.

The custodian must be able to consolidate, receive or send all necessary information as part of its duties. This means possessing control of the smart contracts that issue tokenised assets/tokens, for example, via a multi-signature key. A consequence is that the custodian/asset servicer may need to be involved upfront in the design of smart contracts that issue assets or funds, or develop comfort via appropriate smart contract audits post the design.

This follows on from and reflects the changes introduced by the use of DLT as a technology enabler, to re-imagine funds and fund operations. Ultimately, the self-execution of income, corporate actions and settlements through smart contracts will eliminate a major fraction of asset servicing workload.

5 Operating models for tokenised funds

The premise of tokenising a fund and the use of DLT is to enhance, transform the form of the fund share register and/or change the governance of (and therefore access to) the fund share register.

5.1 Archetypes of tokenised funds

This section discusses common archetypes of fund share registers. The range of archetypes enable the delivery of various value propositions for end investors and expand target market reach for fund managers and distributors.

5.1.1 Archetypes of tokenised fund share registers forms

In practice, there are three general archetypes under which the form of fund share register changes through "tokenisation".

The form of the fund share register selected for a particular tokenisation model is driven among other things by certain value propositions (e.g. cost efficiency versus business model transformation) pursued by the fund manager, distributors, target end investors and service providers.

The ability to convert the form of the fund share register towards a more digital native model, in part is also constrained by the legal and regulatory environment surrounding the fund (e.g. fund domicile driven requirements regarding the permitted registrar(s) such as if digital bearer shares are permitted) and can include associated licensing requirements surrounding the conduct of transfer agency activity.

Model 1	Model 2	Model 3
Mirror the authoritative fund shareholder register	Form part of the authoritative fund share register	Represent entirely, the authoritative fund share register
On-chain copy using DLT Off-chain records using databases	On-chain records using DLT Off-chain record using databases	On-chain records using DLT Off-chain copy using databases

Gold = where the authoritative fund share register is held
Blue = where the copies (mirror/twin) of the share register is held

Table 6: General archetypes of tokenised fund share registers forms

Model 1: Mirroring the authoritative fund shareholder register

This model combines DLT with the authoritative off-chain fund share register, similar to a digital twin model. The on-chain DLT records reflect or mirror changes made in the off-chain records where the authoritative fund share register is held, which means that changes made directly the on-chain DLT records are not immediately (legally) authoritative. The technical implementation of this model would fall under asset type T1[4] – non-native tokens or tokenised representation of assets in the Project Guardian – Enabling Open and Interoperable Networks (2023) paper.

The aim of tokenisation for Model 1 is to enhance cost efficiency through mutualising and simplifying administrative, operational and compliance functions that support investment activities across the

[4] Link to Project Guardian – Enabling Open and Interoperable Networks (2023) paper.

fund value chain. This preserves existing market structures, business and risk operating models, while adhering to existing legal, regulatory and licensing requirements. This model allows fund managers to adopt DLT without having to overhaul existing processes.

Model 2 and 3: Partial or full on-chain authoritative fund shareholder register

In these models, tokenising the fund share register involves the creation of a token that is the source of truth representing, either partially or entirely, the authoritative record of fund shares on a blockchain or distributed ledger. The technical implementation of this model will fall under asset type T2 – native tokens in the Project Guardian – Enabling Open and Interoperable Networks (2023) paper.

Models 2 and 3 introduces an opportunity for new market structures, business and risk operating models, which will impact legal, regulatory and licensing considerations. This arises as changes made to the share register are immediately authoritative, alongside the possibility for multiple verified market participants to asynchronously access and update on-chain share register. These models come with new and novel risks but also allows for the creation of new value propositions via new types of value transfer patterns and mechanics, which are not as easily implemented or scaled through existing infrastructure.

As more of the authoritative fund share register is brought "on-chain", the more that activities performed by existing fund service providers can be reorganised and/or streamlined. Activities historically performed as outsourced services to fund custodians/bank accounts (e.g. fund related payments), fund administrators (e.g. NAV or "proof of reserve" calculations), and transfer agents (e.g. managing share registry access and updates vis-à-vis coordinating subscription, redemption, transfers), may now:

- see new types of market participants and technology service providers like node operators, block explorers and analytics solutions, payment services providers, blockchain risk analytics solutions, data oracles, and wallet providers.
- be represented and performed autonomously by DLT itself (e.g. the operational processing performed inherently by the self-synchronising shared ledger and smart contracts).
- be in-housed by financial intermediaries in the fund value chain, like fund managers, distributors or end investors themselves.

5.1.2 "Governance Archetype" of tokenised fund share registers

Closely inter-related to "Form Archetypes", are "Governance Archetypes" of tokenised fund registers. As part of industry practices today to diversify risks tied to specific financial institutions, many end investors hold value (i.e. assets, cash) across different banks, custodians and service providers. In a tokenised world, these assets in their tokenised forms will likely be held across different DLT networks (e.g. from public-permissionless to private-permissioned). This would necessitate the design of the fund register tokens to be interoperable across such networks for investor access.

Depending on the DLT network properties, the governance archetypes of the tokenised share register will need to be structured accordingly to ensure compliance with relevant laws and regulations governing any fund. There are two forms of permissioning that can be implemented, either on the share register token itself, or on the ledger where the share register token exists.

Implementation of key governance controls	
Ledger-based permissioning (typically applied in private-permission style ledger networks or public permissioned (Layer 1-2) networks)	Smart contract-based permissioning (typically applied in public-permissionless style ledger networks)
Ledger-level permissioning	Access via whitelisted Node / Web3 wallets interface Smart Contract-based permissioning

Blue = where permissioning is implemented in the DLT stack including wallet
Gold = where permissioning is via the interface that users connect to the tokenised fund share register

Table 7: Governance archetypes of tokenised fund share registers

Referencing the use cases in Project Guardian[5], different tokenisation archetypes based on value proposition and target market can be broadly classified in the following categories:

Smart-contract based permissioning	- UBS AM tokenised VCC - Franklin Templeton Investments tokenised VCC
Ledger-based permissioning	- Schroders eVCC use case - Deutsche Bank DAMA2 multi-chain asset servicing

Table 8: Examples of archetypes deployed by financial institutions in practice

5.2 Data models

Regardless of the tokenisation arrangement, fund managers, distributors, and their service providers can elect to incorporate different data types into the arrangement that can be broadly categorised into two categories:

Exogenous data	Endogenous data
Data generated by a trusted counterparty off-chain and published on-chain directly or in an anonymised way	Data generated naturally through operation of the DLT arrangement itself
Examples for funds: NAV prices, portfolio holdings, subscription / redemption details, wallet risk scores	**Examples for funds:** Fund share balances, transactional data (transfers, subscription and redemption)
Data begins off-chain, and can remain off-chain or be brought on-chain (e.g. via data oracle)	Data begins on-chain, and can be brought off-chain as required (e.g. feeding downstream internal fund accounting, risk/reporting systems, off-chain share holder register copies)
Risk considerations: The data workflow from its creation to publication on chain is subject to traditional risk and controls associated with model risk	**Risk considerations:** Subject to newer risks and controls unique to, among other things, the governance and consensus mechanism of the underlying Ledger

[5] Link to MAS Project Guardian website.

Exogenous data	Endogenous data
management and robustness and integrity of transmission of data across IT environments.	impacting settlement finality (e.g. when the blockchain record state can be assumed to be 'final') and ability to synchronise blockchain data for use in downstream risk and operational processes.

Table 9: Possible data models

For fund managers, the design choices around what should be made available on-chain vs off-chain is driven by requirements including those set by business, target technology stack and compliance, data and information security teams, as well as go-to-market timeliness and cost effectiveness.

Generally, the more data that is available on-chain, the greater the ability for fund managers and distributors to pursue new business and operational models by allowing participants to interact on-chain with the data and compose different value propositions (business model). Additionally, it supports the use of analytics to manage, monitor and mitigate compliance related risks (compliance) but at the trade-off of more nuanced client and personal data considerations depending on the tokenisation arrangements/design (data and information security).

6 Guardian Composable Token Taxonomy

Even as the adoption of asset tokenisation gains momentum within the financial sector, many tokenised assets remain siloed and bound to the underlying infrastructure provided by the issuing financial institutions. The introduction of common technical standards enables compatible digital assets to be directly exchanged and moved across DLT networks.

However, realising this in practice is non-trivial.

- **Need for holistic approach**. Standards are inherently shaped by the perspectives and expertise of those who draft them even when they address the same topic. For example, compliance professionals may focus on regulatory adherence and legal frameworks with stringent oversight requirements while in contrast, technologists may emphasise the security protocols and business users would prioritise customer experience aspects. There is no widely accepted approach for the creation of tokens. Token standards in its current form are largely generic and exist well below the level of specific asset class attributes. While there are emerging efforts at specific asset classes to standardise how tokenised assets are represented (e.g. ICMA's Bond Data Taxonomy), the implementation across other asset classes is uneven.

- **Limited Commonality.** As described in Table 10, there is limited commonality across fund products and asset classes. Existing laws, regulations and data apply to a particular asset class or fund type. Each new class and product, requires a new build. Consequently, it takes a long time to launch a new product or asset class and service it with core, robust data and technology.

Differences across asset classes
There is a wide variety of title indicators deployed as markers of ownership across asset classes. Different asset classes frequently have very different forms of title. For example, • Real estate has title deeds; • Equities have shares; • Bonds have indentures and certificates; • Trusts have trust deeds; • Money has bank balances and bearer cash; • Private funds have limited partner agreements; • Unitised funds have units; and • OTC derivatives have ISDAs / CSAs. The attributes and behaviours of each of these title forms are defined in distinct property law. Each asset class also has its own operating / trading models and regulations. Often, unique technology platforms and data are required too, to represent and implement the distinct class-specific models, regulations and laws that are associated with each asset class. Technical standards need to be able to reflect such characteristics.

Table 10: Differences across asset classes

The **Guardian Composable Token Taxonomy (GCTT)**, developed by Project Guardian's asset and wealth management industry group seek to address these limitations through the development of a future state architecture that is flexible and reflect distinct and different asset class-specific requirements, definitions, rules, titles and data.

GCTT builds upon the work of other Guardian industry working groups. For example, the Guardian Fixed Income Framework provides an industry guide to implementing tokenisation in Debt Capital Markets.

With an ecosystem of compatible tokenised assets established, functional components that operate independently may be brought together. By designing tokenised assets as composable modules, new tokenised assets, which are a composite of multiple asset classes can be readily created. This gives fund managers the ability to provide investors with more customised investment options.

6.1 Guardian Composable Token Taxonomy

GCTT comprises four levels of abstractions, with each being composed from reusable layers of lower-level definitions. These levels are defined by the minimum level of functionalities rather than specifically to smart contract protocols to support business requirements and use cases.

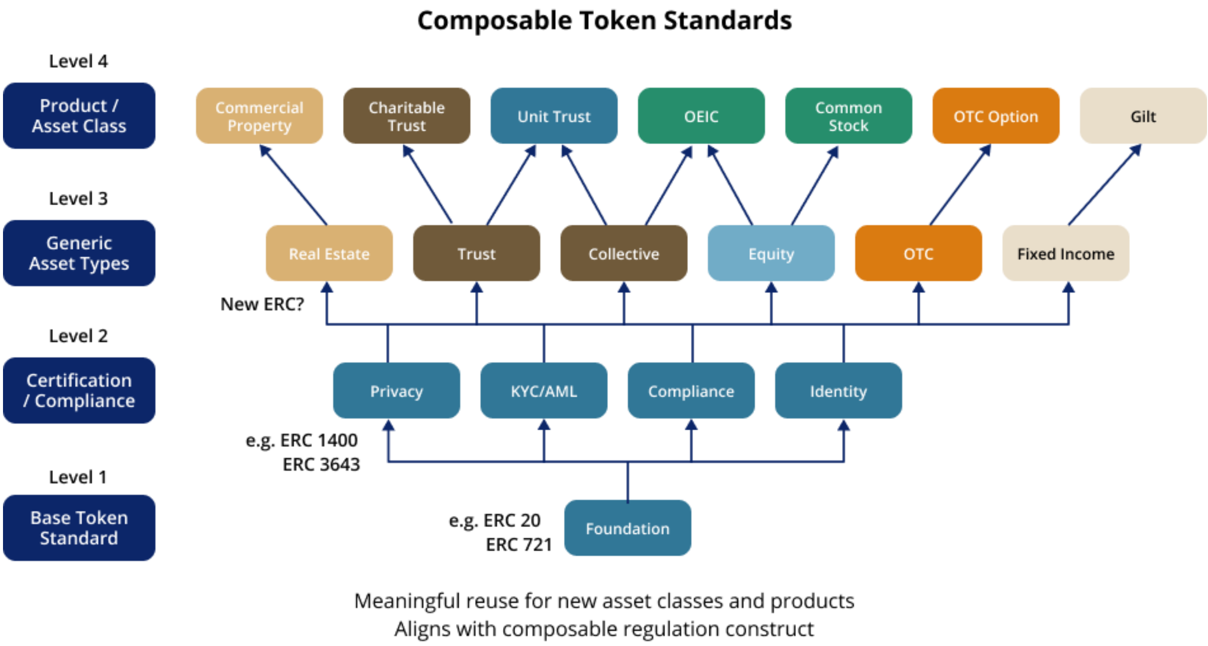

Figure 5: Abstraction levels of the Guardian Composable Token Taxonomy (GCTT)

From the lowest level up, these are:

- **Level 1: The foundation level.** This level sets out the base token standard (i.e. ERC-20, ERC-721), with the characteristics of fungible and non-fungible tokens to be minted, issued and transferred;
- **Level 2: The certification and compliance level.** This level adds conditionality, eligibility and accreditations to the token (i.e. ERC-1400, ERC-3643), allowing for whitelisting functionality in the asset token *(Note: this is not the only implementation of whitelisting)*.
- **Level 3: The generic asset-type level.** This level determines the asset type, and the generic information on the underlying asset, ownership, management control of holdings, and such.
- **Level 4: The specific product type level.** This level determines the product type (e.g. financial instrument, investment vehicle, etc.) with specific information (e.g. prospectus, pricing, disclaimers etc.), including the adding of terms and restrictions that distinguish the management of holdings and trades within the individual asset class.

The figure below provides a high-level functionality and technical description of the **GCTT**.

Levels	Functionality	Description
Level 1	Base token standards	Smart contract governance/multi-sig for issuance, transfer, burning; fungible, non-fungible. GCTT draws on ERC-20 for fungible tokens and ERC-721 for non-fungible tokens as reference standards for interoperability.
Level 2	Transaction flow functionality	Identity authentication and verification, conditionality in transfer events, whitelisting, blacklisting, privacy. GCTT draws on ERC-3643 and ERC-1400 as reference standards.
Level 3	Asset and asset servicing functionality	Modelling equities, cash, fixed income, and their lifecycle activities and workflows including asset level servicing.
Level 4	Fund product type functionality	Modelling financial instruments, investment vehicles / fund structure, e.g. unit trusts, company structures, discretionary portfolios and the fund structure's lifecycle activities and workflows including product level servicing.

Table 11: High-level functionality and technical description of the GCTT

The next section articulates this focus with examples to facilitate further discussions and refinement of approach and this standard outline.

6.1.1 Level 1 – Base token standard

The level 1 base token standard supports the creation of digital assets in a fungible (e.g. fund, securities or cash tokens that can be traded like-for-like) or non-fungible form (e.g. identity tokens that are unique and cannot be easily replicated). It draws on the standard functionalities of ERC-20 for fungible tokens and ERC-721 for non-fungible tokens with the following as the minimally required functionalities for business-as-usual activities:

<table>
<tr><th colspan="4">Minimum required functionalities for Level 1 – Base token standard</th></tr>
<tr><td>Issuance/Subscription</td><td>Approval for operator to transfer tokens</td><td>Retrieve holding for a specific holder</td><td rowspan="2">Link token to metadata with uniform resource identifiers</td></tr>
<tr><td>Mint/Burn</td><td>Approval from holder to transfer tokens</td><td>Retrieve holders of a specific token</td></tr>
<tr><td>Redemption</td><td>Holder transfer of tokens to Recipient</td><td rowspan="2">Multi-signature arrangements</td><td rowspan="2">Retrieve token name, symbol, supply and other token-related information</td></tr>
<tr><td>Event Logging</td><td>Recipient claim of tokens from holder</td></tr>
</table>

Table 12: Minimum functionalities for GCTT Level 1

6.1.2 Level 2 – Certification and compliance

Level 2 focuses on the certification of identity before allowing for transaction level actions such as cross-network token interactions or transfers[6], on-off chain interactions and uniform resource identifiers (URI), and compliance related standards.

It draws on the standard functionalities of ERC-3643 and ERC-1400 with the following as the minimally required functionalities to ensure compliance is in adherence:

Minimum required functionalities for Level 2 – Certification and compliance		
Holder on/off boarding	Verification of holder	Transaction approval
Digital/non-digital identity identifiers	Access rights and control – issuer	Transaction rejection
Whitelisting of wallet addresses for interaction	Visibility of confidential data	Track transaction statuses
Cross-jurisdictional compliance rules	Freeze / Force transfer (burn) of tokens	Track transaction rejection statuses

Table 13: Minimum functionalities for GCTT Level 2

6.1.3 Level 3 – Generic asset types

Level 3 covers the generic asset types underlying a tokenised fund. The underlying asset can take the form of traditional or digital assets. The functionality of level 3 will need to ensure asset lifecycle events including corporate actions and entitlement dates can be executed. Thus, this layer enables the custodian of the fund to clearly demonstrate control and custody of such assets that could exist on different networks.

The minimum functionalities of this level will be asset class specific and rely on asset class standards, for example – fixed income standards such as the International Capital Market Association (ICMA) primary market handbook[7] for the issuance of debt securities in the primary market. The figure below highlights the minimally required functionalities from a fund's perspective.

Minimum required functionalities for Level 3 – Generic asset types		
Subscription management	Subscription / Redemption fee calculation and payment	Data price feed for asset pricing
Redemption management	Request queue / Order management	Expense capture for NAV calculation
Cash management	Performance measurement	NAV Calculation / Pricing
Liquidity fee calculation / payments	Management fee and other fee calculation and payments	Distribution calculation / management

Table 14: Minimum functionalities for GCTT Level 3

6.1.4 Level 4 – Product

Level 4 moves one level deeper from the asset class to the product. Specifically for funds, this level establishes the fund entity type, for example, trust, partnership or corporate, and the separation of the fund vehicle from the underlying assets. Such a separation is achieved via rules driven by specific jurisdictional regulatory compliance attached to the fund entity type. The figure below highlights each major fund type:

[6] Link to Project Guardian – Interlinking Networks Technical Whitepaper (2023).

[7] Reference to ICMA's Primary Market Handbook.

Fund Types		
Principal-traded / open-ended	Market-traded / close-ended	Private / capital commitment-based

Table 15: Major fund types

Level 4 is also envisaged to allow the treatment of the token for collateral eligibility, local taxation and capital adequacy. In this case, whitelisted parties can interact with the token through smart contracts, allowing for self-execution of relevant computations.

The figure below illustrates an example of fund compliance rules that would differentiate fund types and funds domiciled in different jurisdiction.

Eligible Assets		
Equities	Convertibles	Repo / Reverse Repo
Corporate Bonds	Certificates of Deposit	Futures / Forwards
Government Bonds	Commercial Paper	Options
Warrants and Rights	Treasury Bills	Swaps

Concentration / Issuer Quality	Liquidity / Risk
n% of value in secs / MMI issued by single normal entity (<= 10%)	n% liquidity fees at m% liquidity (<= 2%, <=c.10% liq assets)
n% of value in secs / MMI issued by single high quality issuer (<= 10%)	Redemption limit / gate trigger (>=10% daily redemptions)
Holdings >n% do not exceed m% in aggregate (>5%, 40%)	n% of exposure to a single normal OTC cpty (<=5-10%)
n% in secs / MMI / OTC / deposits of a single entity (<= 20%)	n% of OTC exposure to a single regulated credit inst (<=10%)
n% of value in secs / MMI issued by single group entity (<= 20%)	n% of value in deposits with a single credit inst (<=10%)
n% of value in a single industry (<= 25%)	n% of exposure to an issuer in tracked index (<=20%)
n% of value in secs among m% in aggregate (<= 5%, 75%)	n% of exp to m large issuers in tracked index (<=35%, 1)
n% of value in a unlisted securities (<= 10%)	n% of issued capital from one issuer (<=10%)
n% of value in private equity / venture capital funds (<=c.20%)	
n% of value in third party CISs (<= 100%)	
n% of value in CISs regulated by a third party (<= 30%)	
n% of look-through in CIS invested into other CIS (<= 10%)	
n% of value in any single CIS with same regulator (<= 20%)	

Figure 6: Example of fund compliance rules for different fund types at GCTT Level 4

The figure below provides a sample tokenised money market fund template with compliance rules that focus on eligibility, concentration, credit quality and liquidity.

Eligible Assets	
Short-term Government Securities	Bankers' Acceptances
Premium Quality Corporate Debt	Certificates of Deposit
Commercial Paper	Repo / Reverse Repo Agreements
Money Market Instruments	

Concentration / Credit Quality	Liquidity / Risk
n% of value in assets issued by a single normal party (<= 5%)	n months maximum maturity (<=13)
n% of value in assets issued by a high quality issuers (<= 10%)	n days Weighted Average Maturity (<=60days)
Holdings >n% do not exceed m% in aggregate (>5%, 40%)	n days Weighted Average Life (<=120days)
n% of value in assets issued by a single group entity (<= 20%)	n% of value in daily maturity (>=7.5%)
n% of value in aggregate in MMI, deposits or OTC (<= 15%)	n% of value in weekly maturity (>=15%)
n% of value in deposits with a single credit institution (<= 10%)	Redemption gate trigger (<=c.30% wkly liquidity, <=10 biz days in 90)
n% of value in assets in Government Securities (<= 99.5%)	Redemption fees trigger (<=c.10% daily liquidity)
n% in Govt Securities for CNAV / LVNAV MMFs (<= 100%)	Liquidity fees trigger (<=c.30% weekly liquidity)
n relevant issues, m% in any one single issue (>=6%, <-30%)	n% of exposure to a single OTC cpty (<=5%)
n% of value in units of other MMFs or UCITS (<=10%)	n% of exposure to regulated credit insts (<=10%)

Figure 7: Example of tokenised money market fund template with compliance rules

For the purposes of Level 4, the proposed **GCTT** envisages the ability to call or compose tokenised fund operations using the following minimally required functionalities:

Minimum required functionalities for Level 4 – Product	
Asset selection	Fund concentration / Asset quality checks (regulatory, prospectus defined)
Transaction management	Fund liquidity / Risk management checks (regulatory, prospectus defined)
Performance tracking	Redemption gates conditions
Liquidity Stress Testing	Minimum initial / subsequent investment amount
Fund eligibility checks (regulatory, prospectus defined)	Liquidity / Income distribution frequency / Notice / Cut-off

Table 16: Minimum functionalities for GCTT Level 4

6.2 Composable Token Flows

When financial assets become composable, the potential to maximise reuse and reduce the workload associated with new products and assets becomes possible. Based on the construction of financial assets and products from their underlying flows, the creation of self-executing tokens of entitlement makes possible a single self-executing operating model across all financial assets and products. This in turn facilitates the automation of more seamless transaction flows.

The core principle of this approach is the ability of tokens representing (and implementing) entitlements to flows of value, rather than just titles of ownership to conventional assets. This allows for more flexibility in defining asset cases and investment products and allows the needs of the investor to be met in a direct manner.

The digital ecosystem illustrated in the figure below reflects how simplified tokenised flows can represent the entitlements that deliver to client flow requirements, alongside tokens of title to conventional assets.

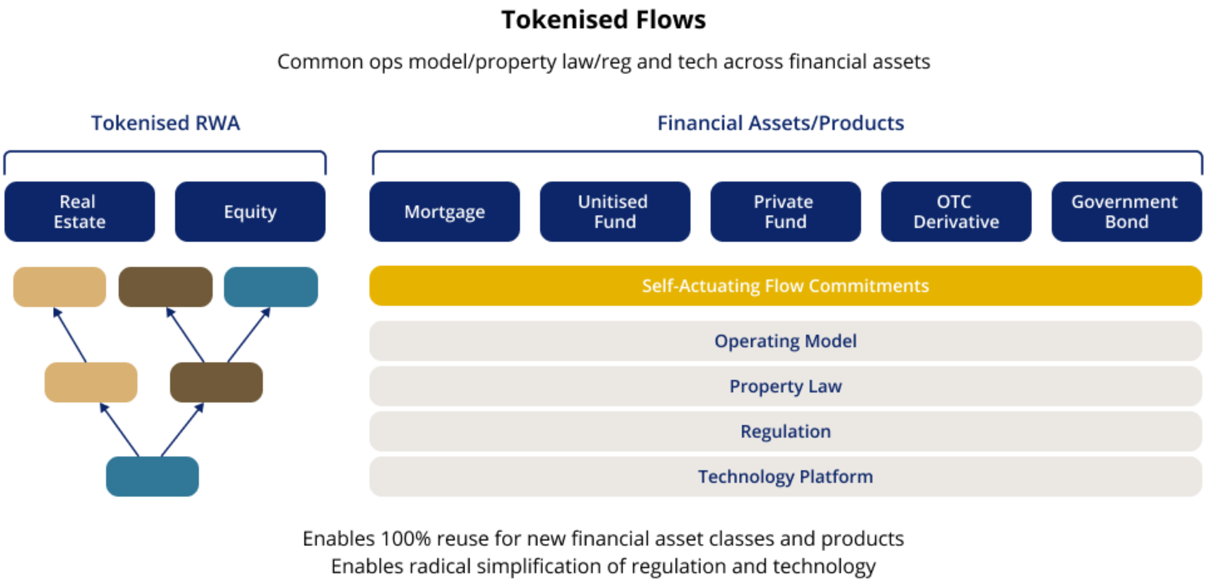

Figure 8: Tokenised flows under GCTT

The role of the asset manager, and the experience of the client, changes significantly. The manager seeks value, and curates a set of inbound tokens and other assets, from which they make flow commitments to their clients. Alongside tokens representing title to non-financial assets, clients will hold tokens representing contractual liabilities.

The liabilities and the flows that they commit may be bespoke to the needs of the individual client or may be drawn from the standard offerings of market participants. For example, a bank offering deposits on a rate curve may satisfy part of the client's requirement. Generally, contingent or non-contingent flows may be offered and accepted along the client's desired outcome curve. The asset manager will manage the assets, and may orchestrate other components of the client solution – but they may not be the source of every commitment.

The committed flows need not be in cash: while cash tokens are expected to be a common form of commitment, the flows committed can be in any form of token. So, while a flow commitment is a contractual liability to the issuer, this does not imply that the issuer always takes risk. There is the opportunity, but not the necessity to risk-share between the issuer and the recipient of a flow commitment token.

The native token issuance approach gives full effect to both current value and future flows, and so delivers directly to the two key components of financial products identified above. In doing so, it delivers directly to our real clients – the investors and borrowers who want to interchange between those current values and future flows.

While this is ambitious and ultimately dependent on substantial change, the composable technical standards and modular approach are incremental steps indicating a pathway towards tokenised asset management.

6.3 Implementation of Guardian Composable Token Taxonomy

This section will cover a sample implementation of the GCTT – ranging from base operations to compliance, asset and asset servicing and fund type. This multi-tiered approach ensures that operations are consistent and reliable while allowing for integration of regulatory and fund prospectus adherence, on-off chain document linking, cross-chain interoperability, specialised asset class and diverse fund structures.

An implementation of GCTT for tokenised fund may feature some of these functionalities as examples:

Fund lifecycle function	Smart contract function	Description of function
Issuance	isIssuable	Checks if new tokens representing fund units can be issued. **Who can call this function?** Fund Manager, Transfer Agent, Approved operator **Inputs**: None **Outputs**: Yes/No
	issue	Creates new tokens and assigns them to a specific investor's wallet address. **Who can call this function?** Fund Manager, Transfer Agent, Approved operator **Inputs**: Investor's wallet address, Number of tokens to issue, Additional data (optional) **Outputs**: Confirmation events (messages)

Fund lifecycle function	Smart contract function	Description of function
Transfer	transferWithData	Moves tokens from the token holder's blockchain address to another blockchain address, with the ability to include extra data. **Who can call this function?** Investor (Token Holder) **Inputs**: Recipient's wallet address, Number of tokens to transfer, Additional data (optional) **Outputs**: Confirmation events and updating of the blockchain ledger (messages)
	delegateTransferWithData	Allows an approved third party to move tokens from one address to another address on behalf of the token holder. **Who can call this function?** Custodian, Broker, or Other Approved Entity **Inputs**: Token holder's wallet address, Recipient's wallet address, Number of tokens to transfer, Additional data (optional) **Outputs**: Confirmation events and updating of the blockchain ledger (messages)
	isTransferApproved	Verifies if a proposed token transfer is allowed based on the fund's rules and regulations. **Who can call this function?** Transfer Agent, Approved Operator, Fund Manager **Inputs**: Recipient's wallet address, Number of tokens to transfer, Additional data (optional) **Outputs**: Approved: Yes/No, Reason for decision
	isDelegateTransferApproved	Checks if a proposed third-party token transfer is permitted according to the fund's rules and regulations. **Who can call this function?** Transfer Agent, Approved Operator, Fund Manager **Inputs**: Token holder's wallet address, Recipient’s wallet address, Number of tokens to transfer, Additional data (optional) **Outputs**: Approved: Yes/No, Reason for decision

Fund lifecycle function	Smart contract function	Description of function
Redemption	redeem	Allows a token holder to return their tokens to the fund and receive the equivalent value in the fund's base currency. **Who can call this function?** Investor (Token Holder) **Inputs**: Number of tokens to redeem, Additional data (optional) **Outputs**: Confirmation of events (messages)
	redeemFrom	Enables an approved entity to return tokens from a specified wallet to the fund and receive equivalent value. **Who can call this function?** Custodian, Broker, or Other Approved Entity **Inputs**: Investor's wallet address, Number of tokens to redeem, Additional data (optional) **Outputs**: Confirmation of events (messages)

Table 17: Example implementation of GCTT for tokenised funds

It is essential to highlight that this list is not exhaustive but is an initial iteration to be expanded. This accommodates the necessity for further enhancements and additions for further iterations.

7 Risk considerations

The following risk considerations are developed for the purpose of using DLT, smart contracts and/or Tokens to represent units/shares in a fund structure. The term ‘Token’, for the purposes of use in risk factor description is used broadly and does not prescribe any specific legal definition. As the adoption of tokenisation continues to grow and evolve, these risk considerations are non-exhaustive and will largely depend on the structure of the transaction. Fund operators will need to ensure the considerations are suitably adapted so that the relevant risks are appropriately disclosed for their investors.

Area	Sub-Area	Risk Considerations
Technology	Nascent technology and smart contracts	Risks relating to the malfunction, coding error or unexpected functioning of the platform or smart contracts, which may have adverse consequences on the registration, settlement, and transfer of DLT-based fund products.
		Risks of inherent flaws and limitations, including permissioned networks.
		Risks of technological immutability, such that if processed in error, a transaction to buy or sell a DLT-based fund product cannot be reversed.
	Cybersecurity	Risks may increase over time, due to advancements in cryptographic technologies and techniques.
		Risks of manipulation due to malicious actors in the network.
		Risks of exploits or hacks due to failure to update (or update in a timely fashion) a protocol or network.
	Data integrity and consistency	Risks of forking of, failure of, or disruption to, public DLT networks or blockchains.
		Risks of compromised by malfunctioning nodes or errors in the underlying source codes.
	Data privacy and access	Risks relating to accessibility of digital assets such as due to compromise of a wallet owner’s private key or due to loss of private key.
		Risks of becoming known to the public which person holds the digital wallet associated with a specific public address.
Legal and regulatory	Jurisdiction variability	Laws and regulations in respect of DLT-based fund products being nascent, highly divergent and non-harmonised globally.
	Evolving regulations	DLT technology is subject to a rapidly evolving regulatory landscape (including tax treatment), which might affect the security, privacy, the ability to buy or sell fund products issued using DLT or other regulatory aspects of DLT transactions and trigger changes to, for example, the DLT networks and relevant documentation. Future regulatory changes could expose the DLT network and participants to new costs and expenses, which may impact the ability to achieve its business objectives.

Area	Sub-Area	Risk Considerations
Operational	Payment and settlement	Risks related to the use of tokens/representations of fiat currency for payment and settlement, and the conversion thereof to and from fiat currency.
	Interoperability	Risks related to the dependency on allowing interactions between different platforms.
	Third party service providers	Risks related to failures by third party service providers, such as cloud service providers.
	Liquidity	Risks in relation to the liquidity of DLT-based fund products potentially being adversely affected by a lack of public trust in the underlying technology.
	Jurisdiction restrictions	Risks in relation to the inability to list and admit to trading DLT-based fund products in certain jurisdictions.
	Existing active market	Risks related to the lack of an existing active market for DLT-based fund products.
Disclosures	Role of Intermediaries	The role that key intermediaries play (e.g. operator of the DLT platform, custodian, registrar, etc), including in relation to KYC/AML and sanctions checks.
	Type of DLT platform	The types of DLT platform used (e.g. public or private) for the issuance of the DLT-based fund products and certain key information in relation to the functioning of such platforms.
	Business continuity	Transparency on business continuity plans.
	Environmental Impact	Environmental footprint concerns of underlying technology.

Table 18: Risk considerations

8 Case study and example

8.1 Tokenised Money Market Fund (MMF) Use Case

Background

Money Market Funds, which have long been a cornerstone of short-term, low-risk investments, are now being reimagined through tokenisation. Money market funds are mutual funds that invest in short-term, highly liquid instruments such as Treasury bills, commercial paper, and certificates of deposit. They are popular among investors seeking to preserve capital while earning a modest return, often used as a safe haven in volatile markets. MMFs provide liquidity, safety, and yield, making them an integral part of the financial ecosystem. This part explores the concept of tokenised money market funds, their potential benefits, challenges and the broader implications for financial markets. Tokenisation refers to the process of creating tokens on a blockchain that represent ownership to rights or assets. In the context of money market funds, tokenisation involves issuing digital tokens that represent a share in the underlying fund. These tokens can be bought, sold, and traded on DLT and non-DLT platforms, offering a new level of accessibility and efficiency.

The regulatory landscape for tokenised assets is still evolving. Different jurisdictions have different rules, and the lack of a global standard can create legal and compliance challenges for fund issuers and investors. Other risks owing to reliance on technology such as increased operational risks including the potential for bugs in smart contracts or failures in the underlying DLT infrastructure needs to be considered. Applicable risks considerations are covered in Section 7.

Approach

The fund manager explores issuance of money market fund in tokenised format, for the purposes of exploring:

- New distribution channels and venues, including newly regulated digital exchanges;
- Use of the tokens as collateral; and
- Potentially create a secondary market layer for the tokens that does not lead to a fund subscription/redemption order each time the token changes hands in a secondary market layer.

This use case does not explore efficiencies in the servicing of the fund itself, which has already been deliberated earlier in this report through Tokenised Investment Vehicles. Instead, this use case purely focuses on the potential that a fund unit as a token explores across the three described areas.

Tokenising MMFs can be approached in several ways, each offering distinct advantages, challenges, and levels of DLT integration. The following section outlines the prominent approaches including their key features.

- **Last mile tokenisation:** The tokenisation process occurs at the final stage of the MMF lifecycle. The underlying assets of the funds are managed traditionally, and only at the point of investor subscription or redemption are tokens issued or burned to represent ownership. The Transfer Agent Register (of Unitholders) can either be fully on-chain (Token holders) or a hybrid of on-chain/off-chain registers.
    (a) Fund Servicing: Traditional fund servicing occurs off-chain using existing processes where the core fund operations (e.g., NAV calculation) are managed.
    (b) Tokenisation at distribution: Tokens are only issued at the final stage representing an investor’s share in the fund upon subscription and burned during redemption.

(c) Minimal disruption: This approach allows for minimal changes to existing fund structures and operations, reducing implementation complexity.

- **End-to-end Tokenisation:** This method integrates DLT throughout more parts of the lifecycle of the MMF, from fund creation and asset management to token issuance and redemption. Aspect of the fund's operations are managed on-chain through smart contracts.
  (a) Full DLT integration: Tokenisation is embedded in more steps, from asset management to fund servicing.
  (b) Real-time transparency: Investors and regulators may have real-time visibility into the fund's performance, NAV, and asset backing, providing continuous transparency.
  (c) Automation: Smart contracts automate processes like compliance checks, NAV calculations, corporate actions, and fund management, improving operational efficiency.

For the remaining section of this use case exploration, the focus is on the last mile tokenisation approach. This approach might be particularly attractive to financial institutions that wish to retain much of their existing infrastructure while still offering a tokenised product and helps with immediate adoption of tokenisation and builds a foundation for further exploration of benefits. There are other models that exist where the records are only kept on-chain or where the fund is being tokenised beyond the last mile in a collateral network for the specific purpose of being used as collateral.

Setup of the Tokenised Money Market Fund (MMF)

- **Process:** The issuer sets the MMF as a Unit Trust or VCC, and appoints a Trustee and/or Custodian accordingly for the fund where the underlying assets and cash of the fund are safe-kept. The fund administrator computes the daily NAV for the fund. This process is traditional and has no changes, however it is important to define the parameters of tokenisation at this stage.
- **Application of GCTT standards:**

  At this stage, foundational attributes such as ownership and transfer mechanisms are established off chain through the legal and structural setup of the fund. The structure ensures that the tokens, when eventually issued, will represent fractional ownership of the MMF.

  It is essential to clearly define the operating model, roles, and standards for all involved parties, as well as the specifications for tokens. Base token standards apply including delineating the type of tokens, establishing clear roles and approval mechanisms such as multi-signature arrangements. Governance functions, such as the issuance of new tokens or their redemption, are guarded by multi-signature mechanisms, enhancing security and compliance.

  Additionally, reconciliation procedures must be outlined, and thorough smart contract audits conducted by independent auditors, these could pave the path for wider adoption. Data considerations are important at this stage too, including specifying the level of metadata the token should carry, the type of data that will be recorded on-chain for public visibility, and the parameters for event logging with the aim of automation via smart contracts. Event Logging record all token-related events (issuance, transfer, redemption) for transparency and auditability.

  Beyond last mile tokenisation, procedures enabled by functions such as the *checkFundAssetComposition* function could potentially ensure that the fund's portfolio complies with regulatory requirements and the fund's investment mandate thereby automatically checking that the assets backing the tokens meet these requirements before token issuance. The smart contracts governing the tokens could be designed to update in real-time with NAV changes, ensuring that the tokens reflect the current value of the fund. The token's smart contract can also be programmed to handle eligibility rules, redemption gates, and liquidity constraints defined by the fund's regulatory framework.

Distribution of MMF tokens

In this use case, two methods of distribution are being explored. First, distribution is via traditional distributors of a fund. Second is the distribution via newly regulated digital exchanges.

- **Process:** A traditional distributor may choose to maintain tokens in a single omnibus wallet aggregated for their end investor holdings, and maintain segregation as book entries in their traditional account ledger system. In contrast, digital exchanges may enable a wallet for each investor, allowing on-chain movement of tokens between investors.

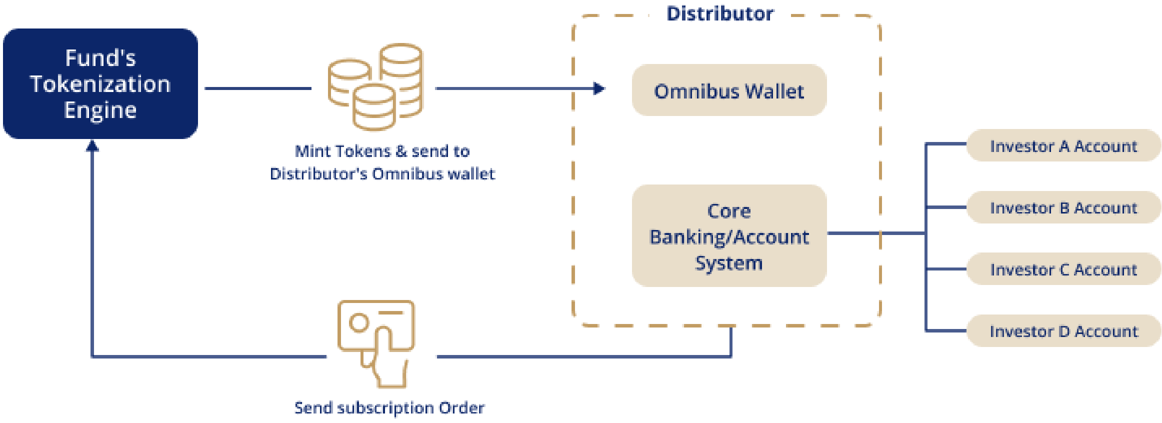

Figure 9: A traditional distributor may receive or maintain all tokens in an omnibus wallet

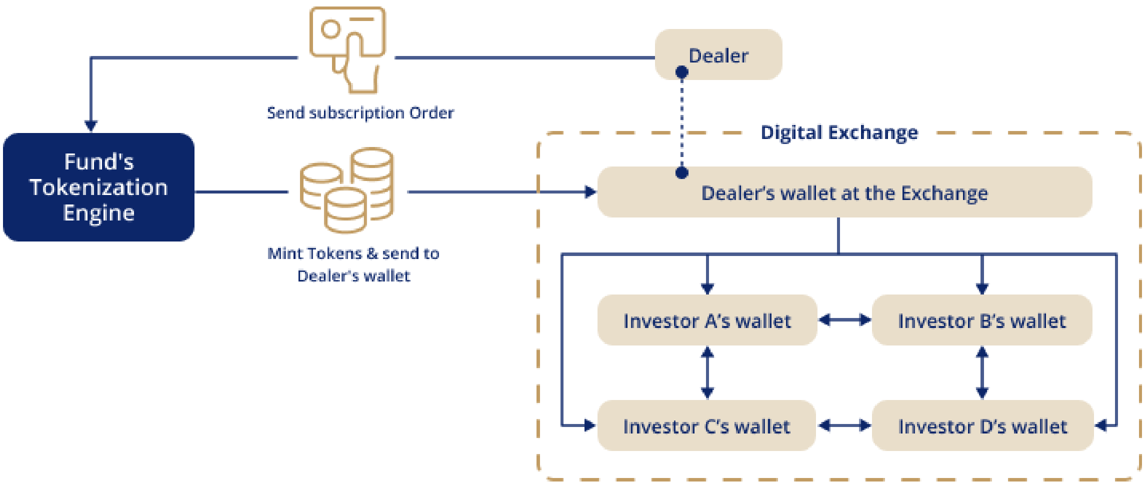

Figure 10: A digital exchange may provide its customers segregated wallets, with tokens moving on-chain between investors

The parties that interact with the Fund's Transfer Agent i.e. distributors and/or dealers, undergo customer due diligence processes and their wallets are required to carry verifiable credentials or be pre-whitelisted for the funds to deposit tokens into these wallets. These credentials are crucial for the security and compliance of the token distribution process. The wallets are whitelisted by the smart contracts governing the fund, ensuring only verified and compliant entities can receive and manage tokens.

The end investors (customers of traditional distributors or digital exchanges) are subject to KYC checks conducted by the distributors. If end investors are given wallets by the digital asset exchange, an operationally effective federated whitelisting mechanism needs to be established to ensure that these wallets carry verifiable credentials.

- **Application of GCTT standards:**

  Emphasising the importance of identity verification, compliance, and secure transfer mechanism, each distributor must register digital identities or verifiable credentials for the wallets they create to ensure secure, transparent operations. The smart contract verifies these credentials before allowing any transactions. By whitelisting wallets and using verifiable credentials, the fund ensures that only authorised and compliant entities can participate in the token distribution process. This also prevents unauthorised access and ensures that all transactions are traceable to the end investor and auditable.

  Functions such as the *verifyIdentity* and *conditionalTransfer* allow the transferor of the fund tokens to ensure only distributor/investor wallet that has undergone KYC and AML checks are receiving the tokens prior to transferring.

  - verifyIdentity – Allows administrators to verify investor (wallet) identity by storing hashes of identity documents, ensuring that only verified investors can participate.
  - conditionalTransfer – Tokens are transferred to whitelisted wallets only if certain conditions are met. This guarantees that the transfer of tokens is in line with AML, other regulatory requirements and ensure authorised participation only.

Subscription and the creation of fund tokens

- **Process:** The subscription process begins when distributors submit subscription orders on behalf of their investors. An orchestrator, acting as the central processing layer, routes these orders to the Transfer Agent. The Transfer Agent is responsible for executing orders, typically done in batches, to ensure efficiency and accuracy. Upon execution, the Transfer Agent confirms the orders and sends contract notes back to the orchestrator. The orchestrator then triggers the tokenisation process by activating the issuance smart contract, which mints the equivalent amount of fund tokens. These tokens are then distributed to the distributor wallets or dealer wallets in the case of distribution via a digital exchange. Any further on-chain or off-chain distribution to the end investors is the responsibility of the distributor or the exchange.

Subscription Flow

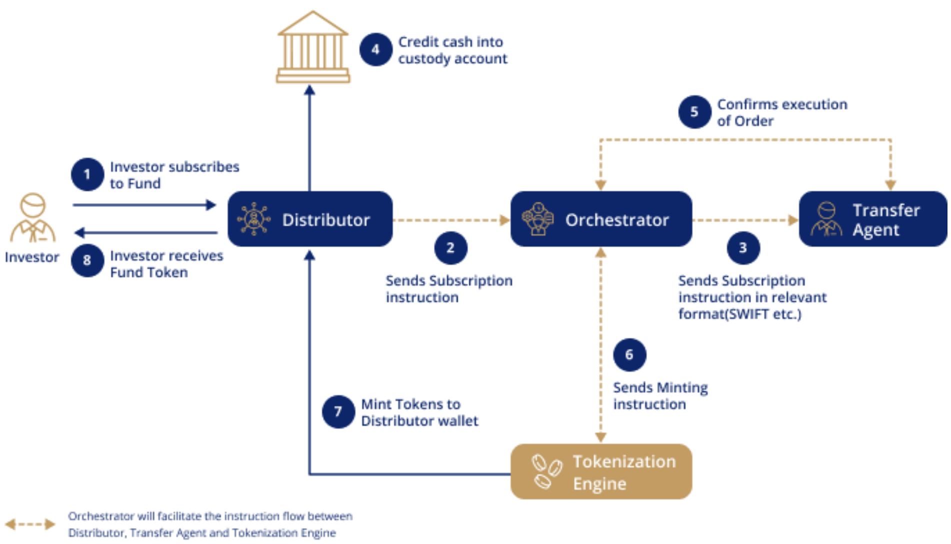

Figure 11: Subscription process

- **Application of GCTT standards:**

  The token standards ensure that each token accurately represent a portion of a fund's assets. The process of issuing tokens is managed to reflect the exact value of the underlying assets, as determined by the NAV calculated by the Fund Administrators. This alignment ensures that the tokens are fully backed by the fund's assets, providing investors with a secure and reliable digital representation of their investment.

  The smart contract mints new tokens in response to confirmed subscriptions.

  - isIssuable: The contract checks whether the fund is still open of new investments before issuing new tokens. It verifies that the token can be created and updates the total supply.
  - mintFundToken: Once authorised, the tokens are minted, and an Issued event is emitted, providing transparency around the new token creation.

  The Transfer Agent's role in executing subscription is embedded in the token's workflow, ensuring that each subscription leads to precise issuance of tokens, calculated based on fund's NAV and the subscribed amount. Event Logging ensures that smart contracts log all subscription events, creating a clear record for reconciliation and audit purposes.

Redemption and burning of fund tokens

- **Process:** The redemption process begins when the investor submits a redemption request through their distributor and returns the tokens to a predetermined wallet for the fund. The orchestrator receives these requests and routes them to the Transfer Agent. The Transfer Agent is responsible for executing redemption order, which includes calculating equivalent value in cash or assets to be returned to the investor. Once the Transfer Agent confirms the execution, the orchestrator triggers the burning of the corresponding tokens via a burn contract. The Tokens are then permanently removed from circulation, reflecting the reduction in the fund's outstanding units. The distributor ensures that the investor received their cash or asset equivalent, completing the redemption process.

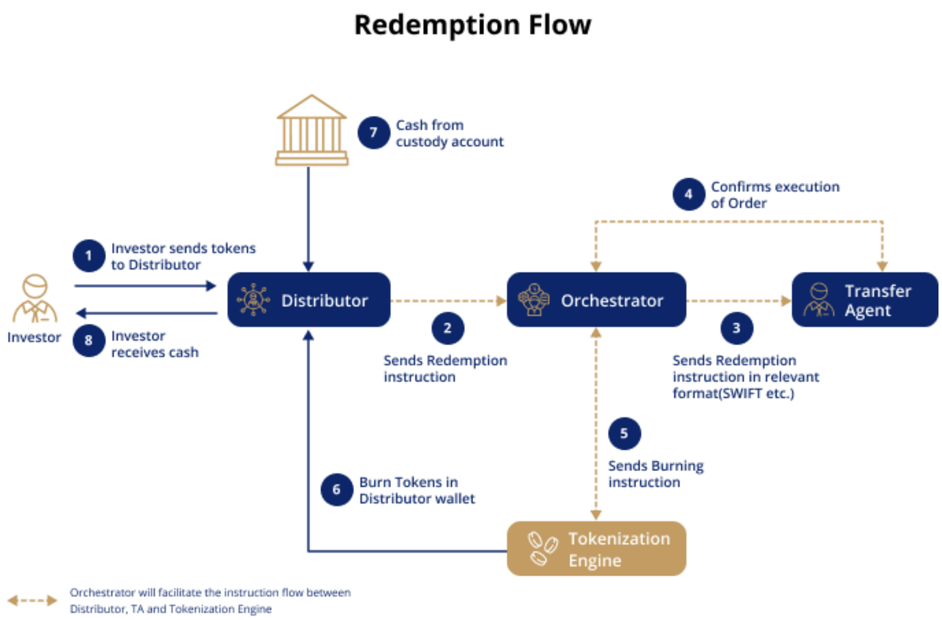

Figure 12: Redemption process

- **Application of GCTT standards:**

  The token burning process follows the same principles as issuance but in reverse. This ensures the total supply of tokens always accurately reflect the fund's outstanding units. The burning mechanism is essential for maintaining the integrity of the token supply and ensuring that each token continues to represent fractional share of the fund. The redemption function reduces the total supply of tokens, and a Redeemed event is emitted to signal the change.

  - Redemption Function - The contract verifies that the tokens can be redeemed, burns them, and records the event.

  The Transfer Agent is responsible for executing the redemption orders, which triggers the smart contract to burn the tokens. The register is updated to reflect the changes in token supply.

  - isRedeemable - The smart contract ensures that the fund is still open for redemptions and that all compliance checks are satisfied before burning tokens.
  - Multisignature Approval - For large redemptions, multisignature arrangements might be used to ensure that all parties approve the transaction before tokens are burned.

  The orchestrator and TA work together to ensure that the redemption process is seamless, with smart contracts handling the burning tokens. This automation reduces the risks of errors and ensures the redemption process is transparent and efficient.

  - dividendDistribution – the redemption process considers whether dividend should be reinvested into the fund. In this case, the investor may receive additional fund tokens through further issuance or have the dividends distributed as cash.

Reconciliation requirements

- **Process:** In addition to existing reconciliation, in this particular use case, additional reconciliation is required_to ensure the tokenised representations of the fund are accurately aligned with the outstanding units.

  Total number of Tokens in circulation = Outstanding Units

  This involves regular checks to verify that the number of tokens in circulation matches the total outstanding units as maintained in the Transfer Agency register. Reconciliation is essential to maintain the integrity of the fund's register and ensuring that all transactions are properly accounted for.

  Reconciliation also involves verifying that all tokens issued, redeemed and burned are accurately recorded on the ledger, with no discrepancies between the on-chain and off-chain records. There could be other models where the register is fully on-chain or the on-chain record becomes a sub-register and there is no requirement for such a reconciliation.

- **Application of GCTT standards:** Smart contracts log all minting, burning, and transfer events, providing a detailed record of token movements. The contract can retrieve the total number of tokens in circulation and holdings of specific wallets, ensuring that reconciliation is accurate.
  - Retrieve Supply Data – Smart contracts allow retrieval of the total supply of tokens and holdings ensuring all transactions are accounted for.
  - Integration with oracles – Pricing oracles provide real-time data on asset values, ensuring that NAV calculations are accurate and reflected in the token's price.

Usage of the MMF Tokens as collateral

The use of MMF shares as collateral has been long constrained because traditional transfer agent (TA) systems are not designed to facilitate the dynamic processes of posting, recalling, and rehypothecating collateral. TA systems, primarily built for recording shareholding and transfer operations, are ill-suited to support the rapid and flexible movements required in modern collateral management. This is one reason MMFs remain "trapped" and underutilised as a collateral asset. Currently, market participants face numerous challenges when attempting to use MMFs for collateral purposes: the systems hinder quick transfers between counterparties, require multiple registrations with various transfer agents, and necessitate MMF redemptions in advance.

Tokenised MMF units on a DLT network fuel their efficient use as collateral and deliver a transformative solution to these longstanding issues. By encumbering MMF units on underlying Transfer Agent ledgers and representing them as programmable tokens on a DLT, this approach enables swift and flexible transfers/pledges of MMF units. This leverages DLT's inherent features such as near-real-time processing and smart contracts, allowing for more dynamic use of MMFs as collateral. It also eliminates any of the current friction points, enabling near-instantaneous transfers and pledges of MMF units between parties without overhauling existing Transfer Agency infrastructure.

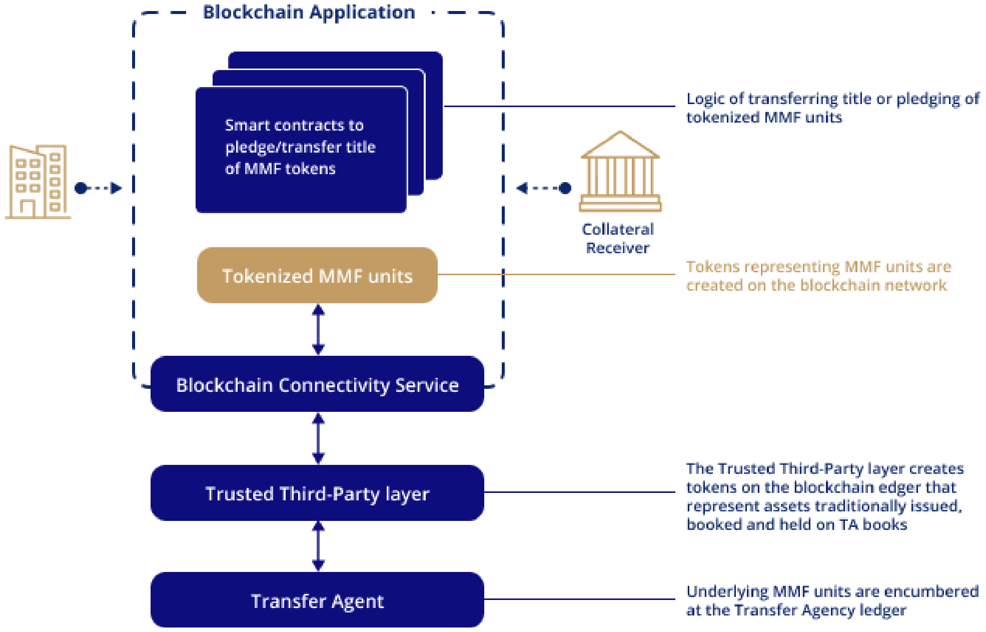

Figure 13: Tokenised MMF units

The use of tokenised MMFs as collateral further introduces a risk management paradigm shift during market volatility. Enabling the direct use of MMF units as collateral margin between consenting counterparties eliminates the need for redemptions (of MMF units into cash), thereby preserving MMF position integrity. This innovation is particularly crucial for asset managers, as it mitigates the risk of forced redemptions and potential drawdowns during market stress. This approach unlocks the latent utility of MMF holdings, transforming them into dynamic collateral assets.

Limitations of the use case

While digital wallets enable direct ownership and management of tokens, adoption of secure custody solutions (i.e., wallet infrastructure and private key management) remain limited among traditional market participants. Digital asset custody is also not offered yet by most traditional Financial Institutions/Custodians. It is recognised that this may rapidly change in the future paving the way for efficient operation of any tokenised asset including funds. Absent this, participants in a fund value chain, may lack the ability to receive or hold tokens presenting a point-in-time challenge in integrating tokenised MMFs into existing financial frameworks.

While the implementation of a DLT solution immediately creates the perception of benefits such as atomic settlements, any instantaneous subscription or redemption of the fund tokens require significant re-designing of traditional processes and workflows, including the movement of cash. The promise of real-time operations in a tokenised MMF may require a combination of quicker deployment of funds, real-time valuation, and instantaneous execution of orders, none of which are easily achievable yet. An instantaneous issuance of fund units can potentially be achieved if the NAV is constant, but redemption will require instant availability of a liquidity pool. A simultaneous issuance/redemption of fund tokens can however be programmatically achieved through receipt of cash and relevant instructions.

Cash clearing and settlement cut-off times in traditional banking systems pose a significant challenge for cross-border transactions in a 24x7 tokenised environment. These cut-off times can delay the movement of funds, complicating the seamless execution of transactions across different time zones. Any secondary market for tokenised MMFs require the presence of dealers and market makers who can maintain liquidity by facilitating the buying and selling of tokens. This is essential for ensuring that tokens can be traded freely, and that process remain stable.

9 Conclusion

This report outlines some important guidance for where value can be achieved and the considerations for charting the path towards realising the benefits of tokenisation. Moving forward on this path is crucial to improve the outcomes for investors and deliver the widest improvements to the financial ecosystem. Tokenisation of funds is therefore the first valuable step; but this is limited without a longer-term view.

Members of Project Guardian's AWM industry working group have piloted and identified benefits from digital assets and are now looking to align standards to scale this commercially. The vision is to bring investment closer and more directly to the investor, maximising efficiency with the deployment of new models and technologies, and thereby reducing costs in the management of investments. There is opportunity to optimise the deployment of DLT and tokenisation to bring real choice and control over investments and outcomes to the investor, and to deliver directly to client needs. This allows funds to evolve from being product focused, to become fully client-driven as an industry.

The approach of composability and token standards in this report represents a transformative shift not only for funds, but the broader financial landscape, to enable greater flexibility, interoperability, and innovation within the investment ecosystem. The principle of composability extends and enables the simplification of legal and regulatory frameworks for investments, with more effective, but simpler controls.

This paper proposed a three-step evolution to deliver to these goals.

1. Tokenised Funds - A standardised approach to the tokenisation of conventional funds (and assets), achieving a level of reuse, without material change from current norms;
2. Tokenised Assets – Using direct holdings in tokenised assets to reduce the costs and challenges of collective vehicles, and giving clients control of their sources of growth and risk; and
3. Tokenised Flows – Adding tokenised entitlements, which enables financial assets and products to become composable, and empowers solutions that deliver directly to client needs.

The proposed end-state vision delivers a radical level of simplification, achieving the objective to maximise the benefit of tokenisation, and benefiting the buy-side and sell-side in equal measure. Significant collaboration will be needed by participants in Project Guardian, and with the wider industry to implement the venues, standards and capabilities that will give effect to this vision.

This initiative welcomes contributions from the broader international community to advance the development of tokenised funds, and to support this transformative shift.

10 Reference

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